

Computational Fluid Dynamics Project Report



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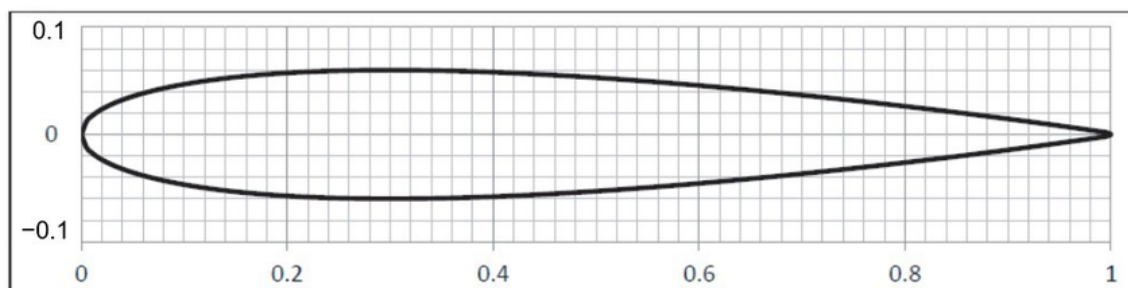
Course:- Fluid Dynamics ME-207

Given:- Freestream velocity - 25 m/s

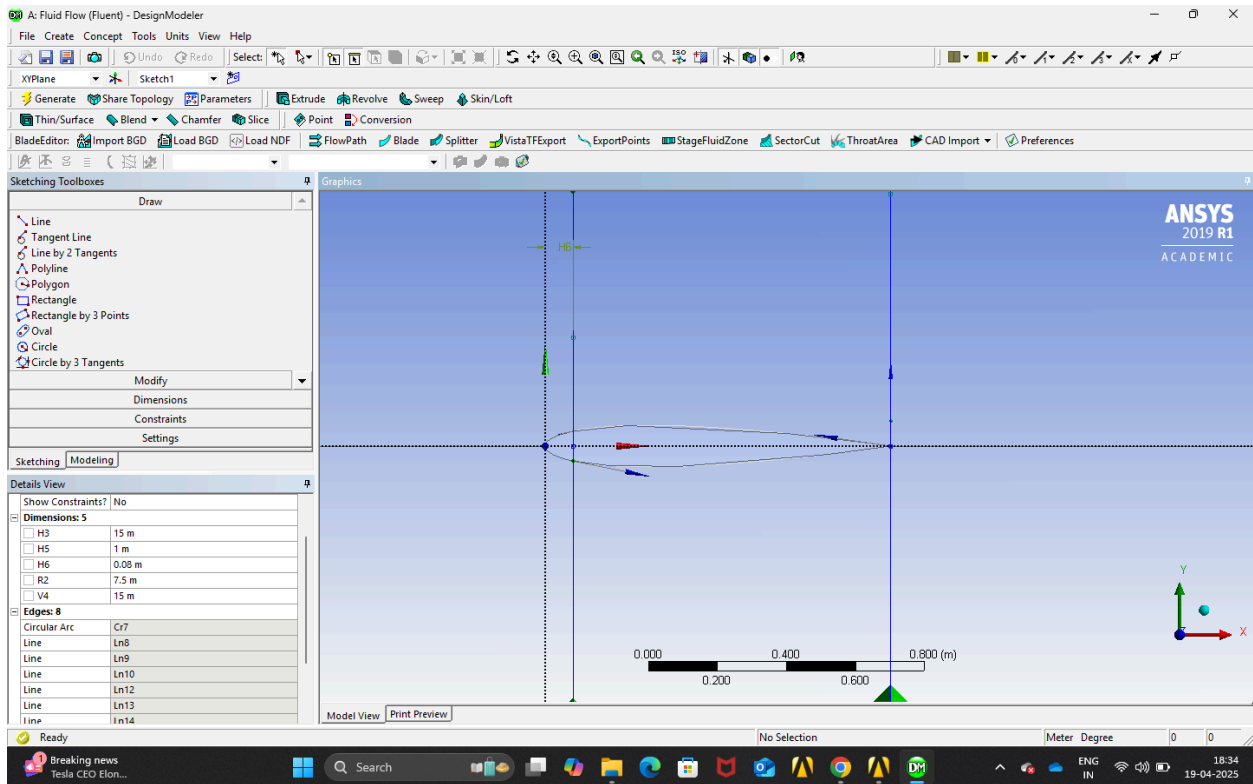
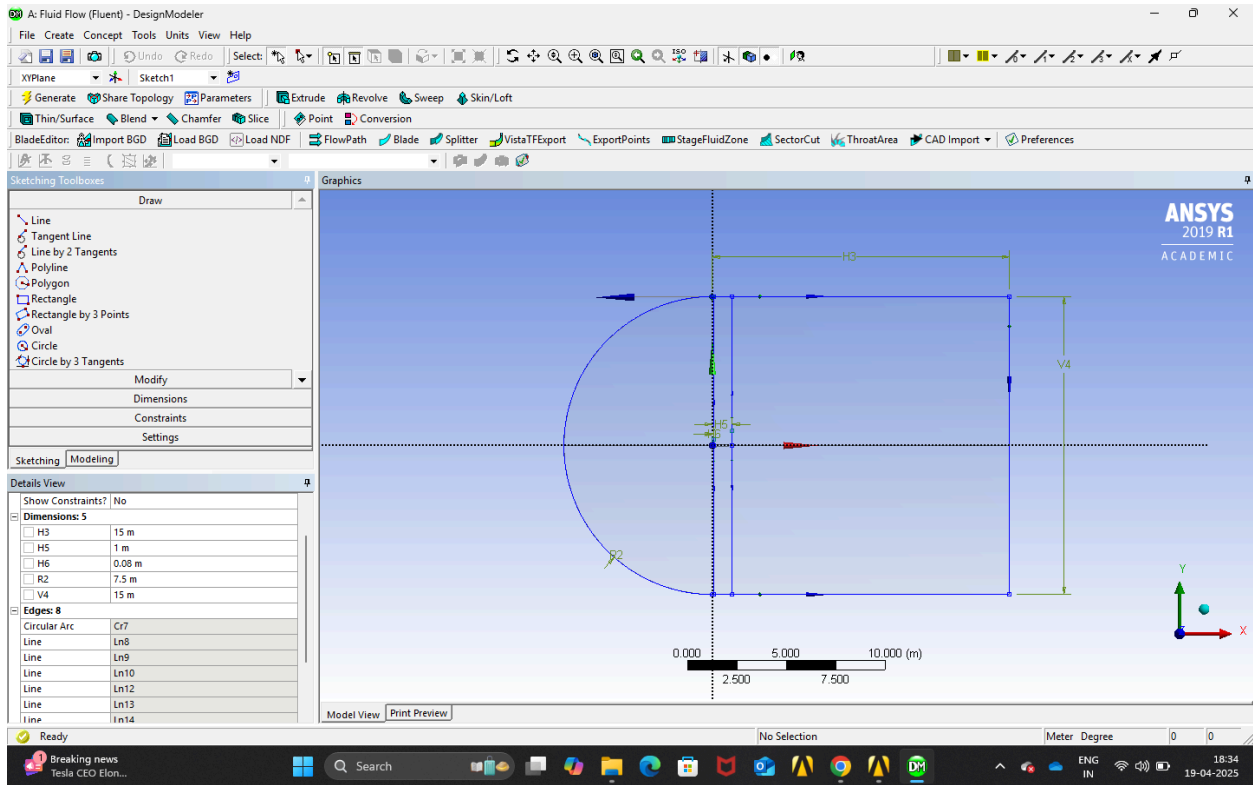
Temperature - 20 degree celsius, Chord Length - 1m

Note:- Simulation and results have been done Ansys 2019 R1 Academic version

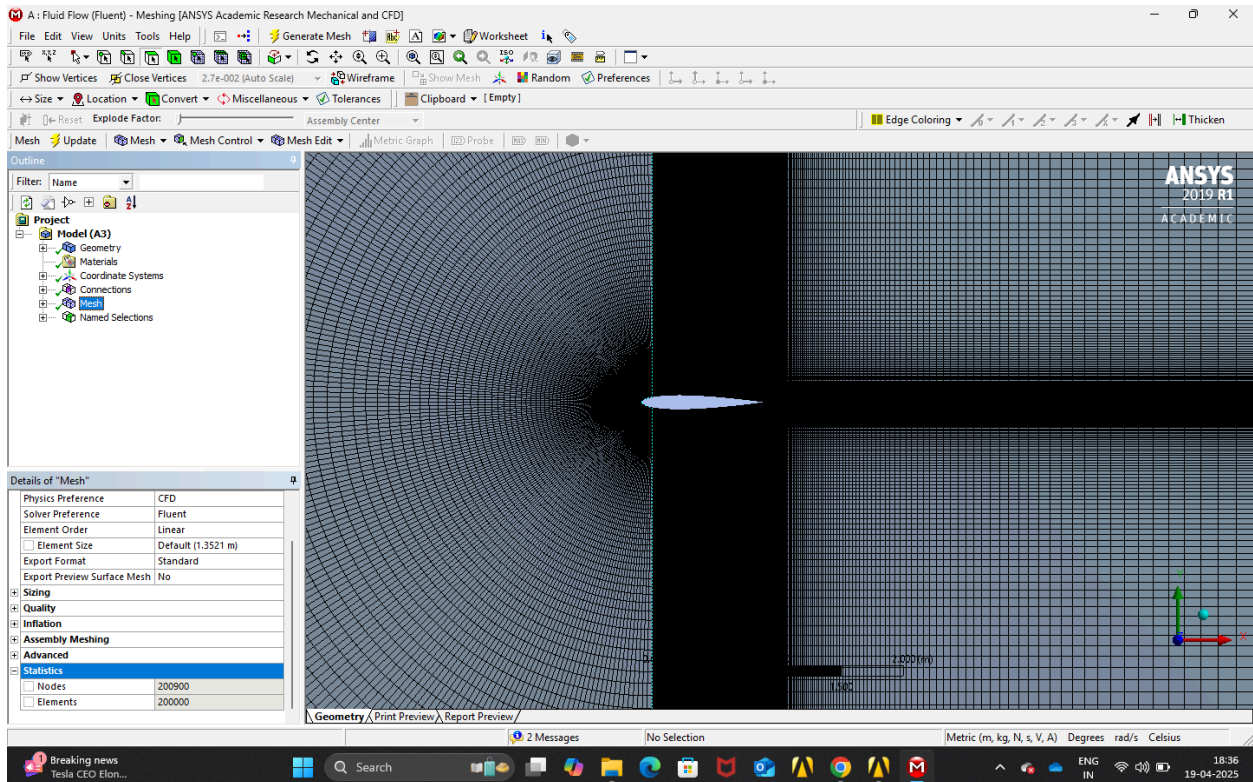
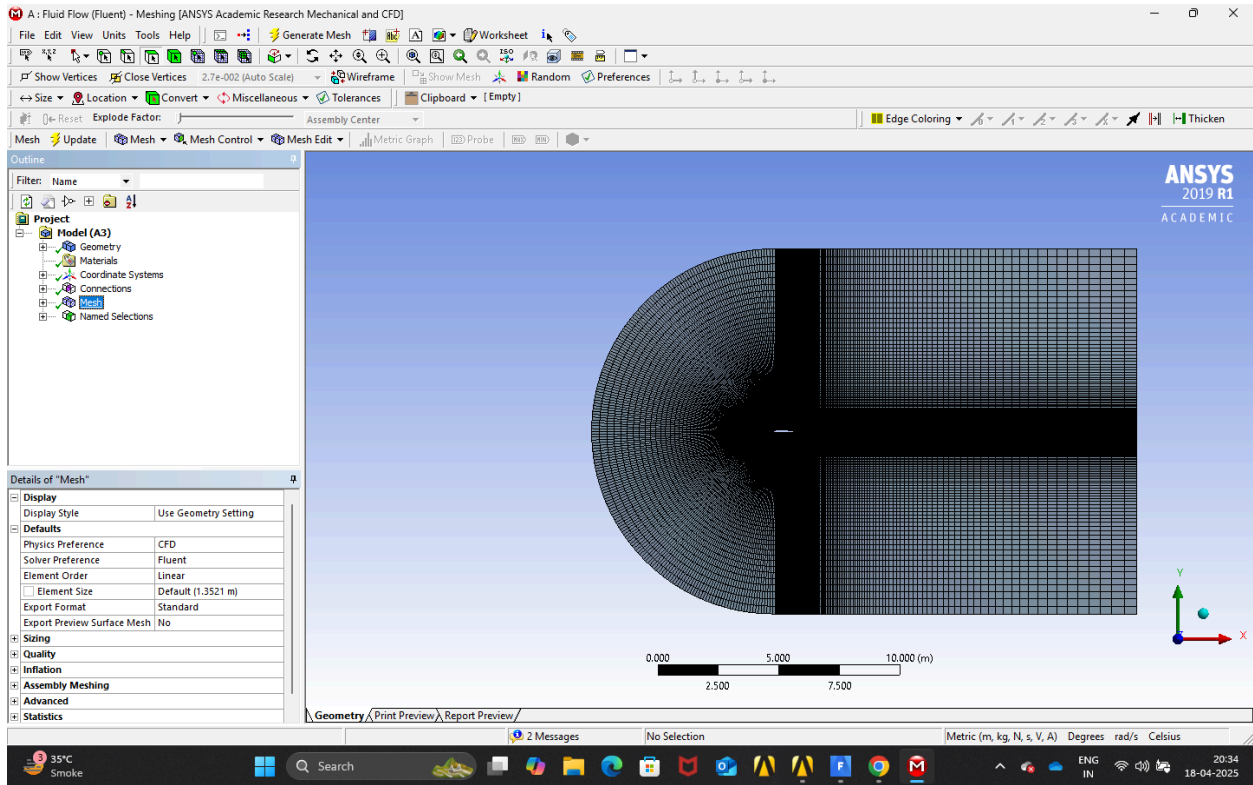
NACA 0012 airfoil section

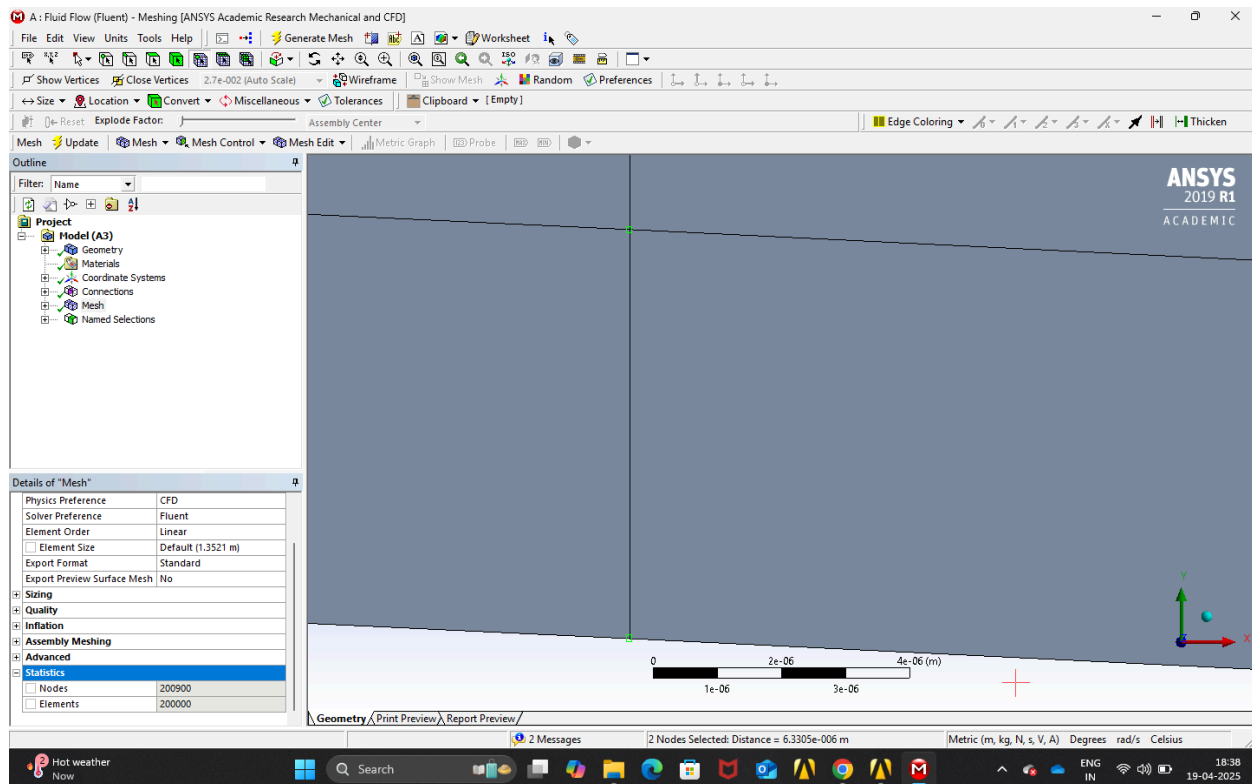


Geometry



Meshing

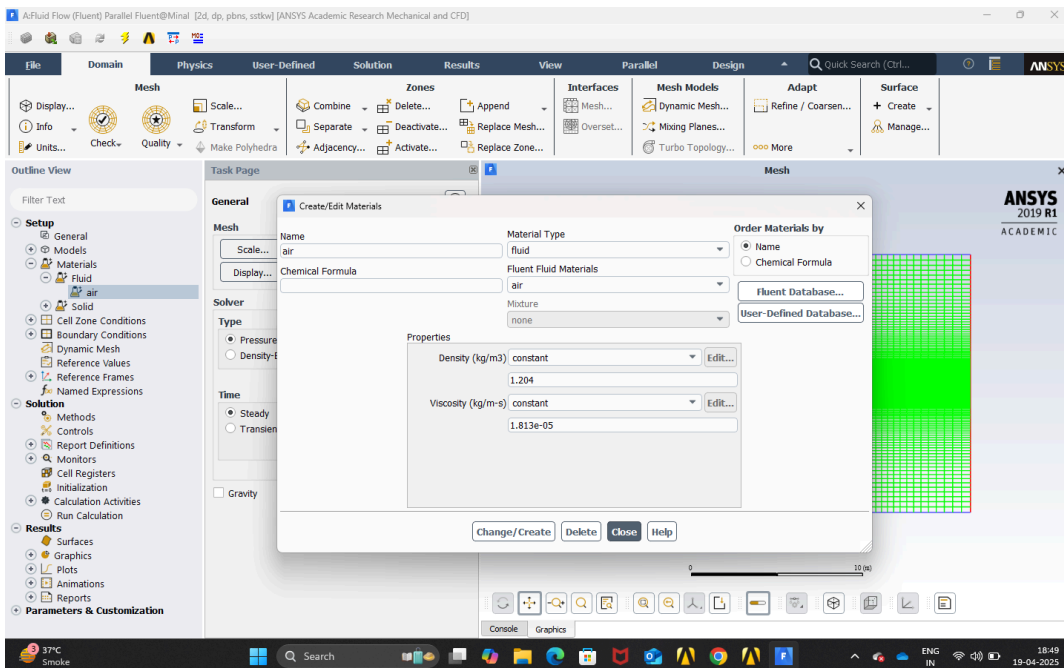
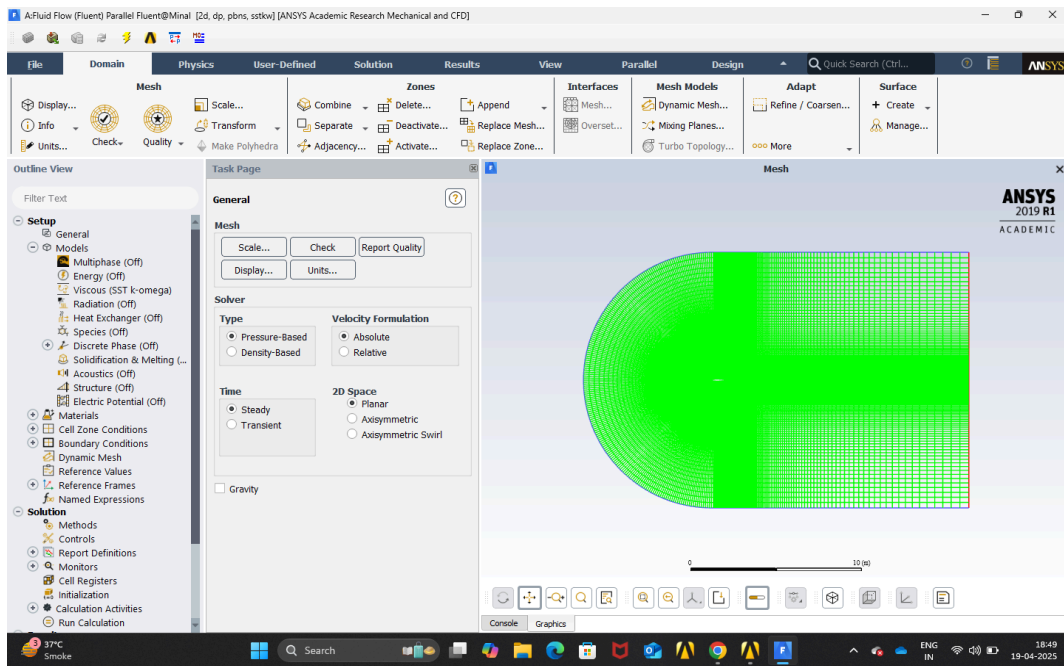




- The total number of nodes = 200900
- The total number of elements = 200000
- The distance between two nodes = 6.3305×10^{-6} m which gives y^+ less than 1, hence the results obtained by the mesh obtained will be accurate and all the boundary effects will be captured accurately.
- The meshing near the airfoil is more concentrated because the airfoil is the region of our interest and if the meshing is not good enough then the results will vary significantly.

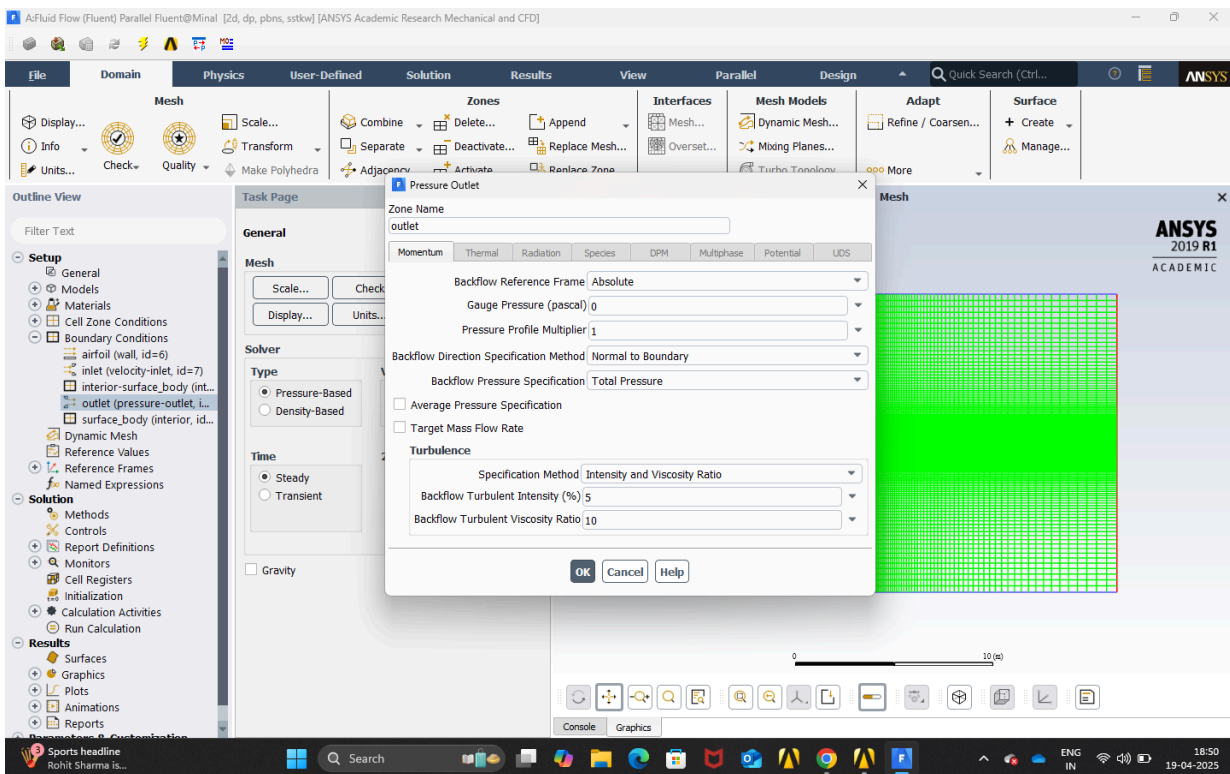
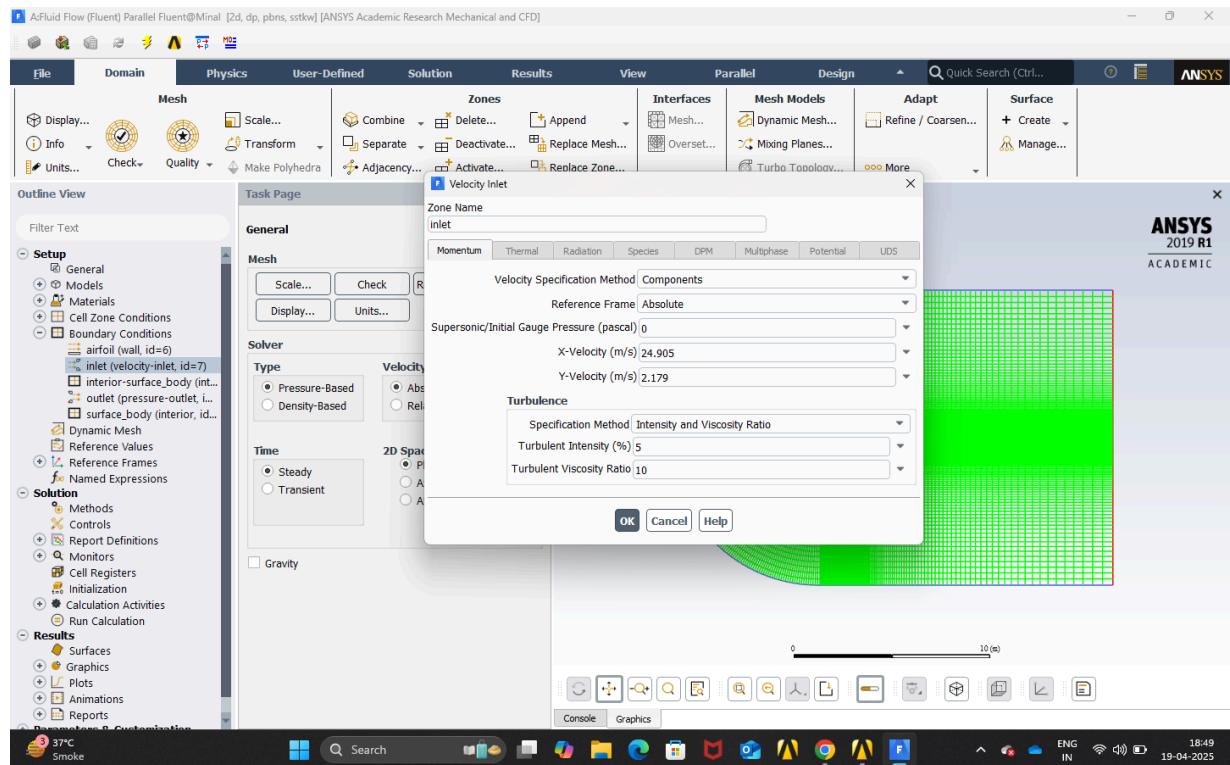
Discretisation schemes and solution methodology (The setup is for Angle of attack of 5°)

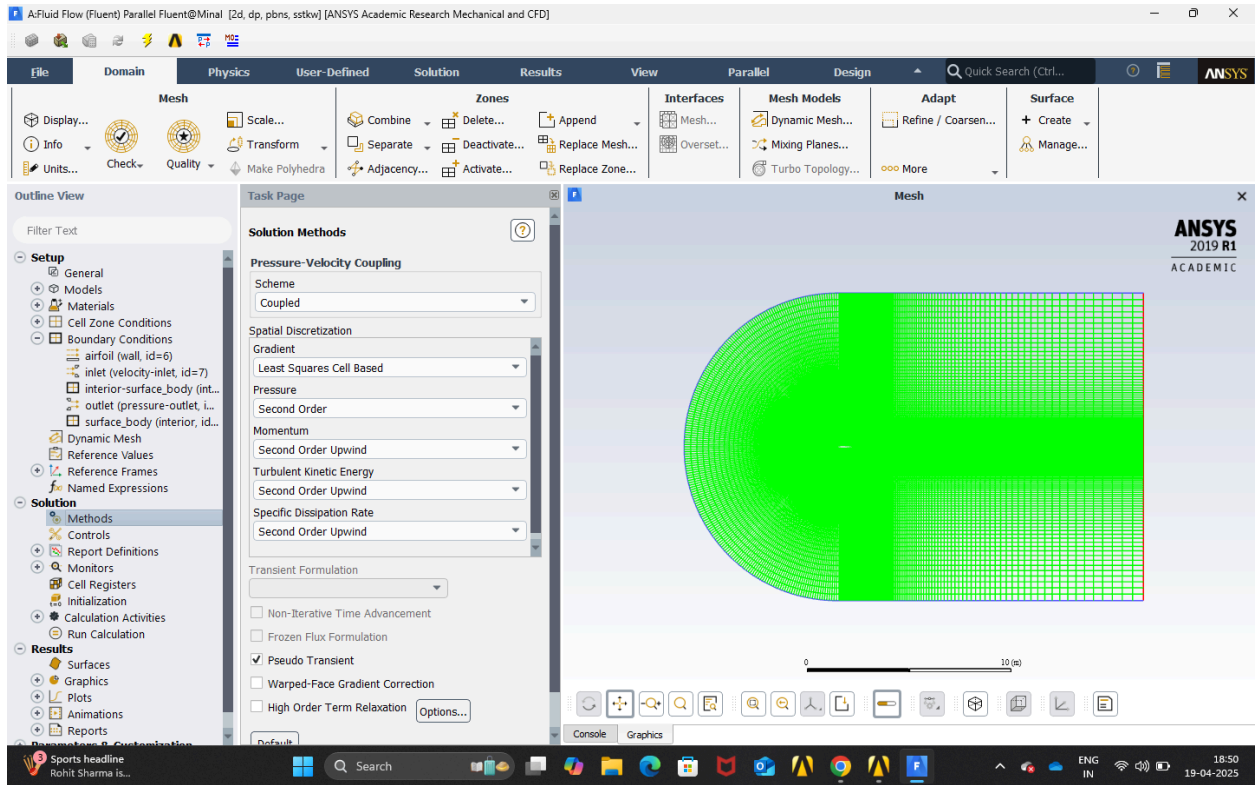
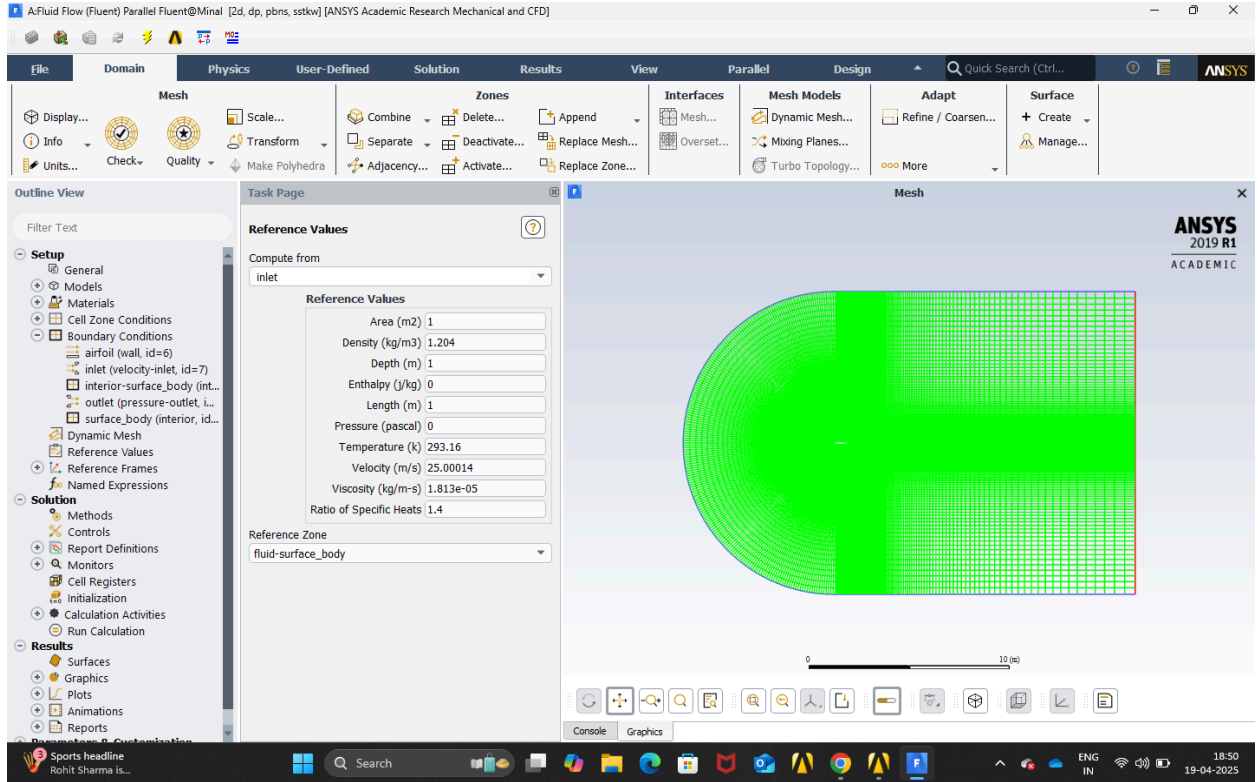
The Reynold's number for the flow is approximately 10^6 so the flow is turbulent as the transitional flow is around Reynolds' number is approximately 10^5 , so the laminar model will not give accurate results so we have to use the SST-K Omega model to get correct results.

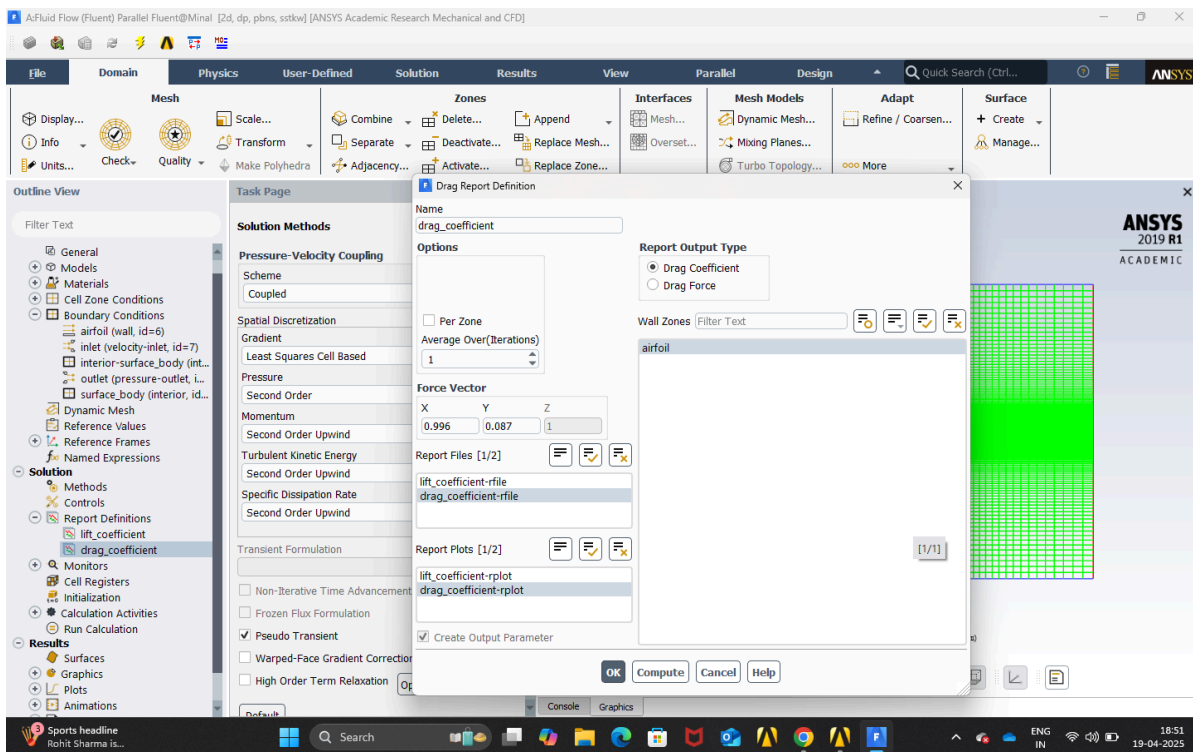
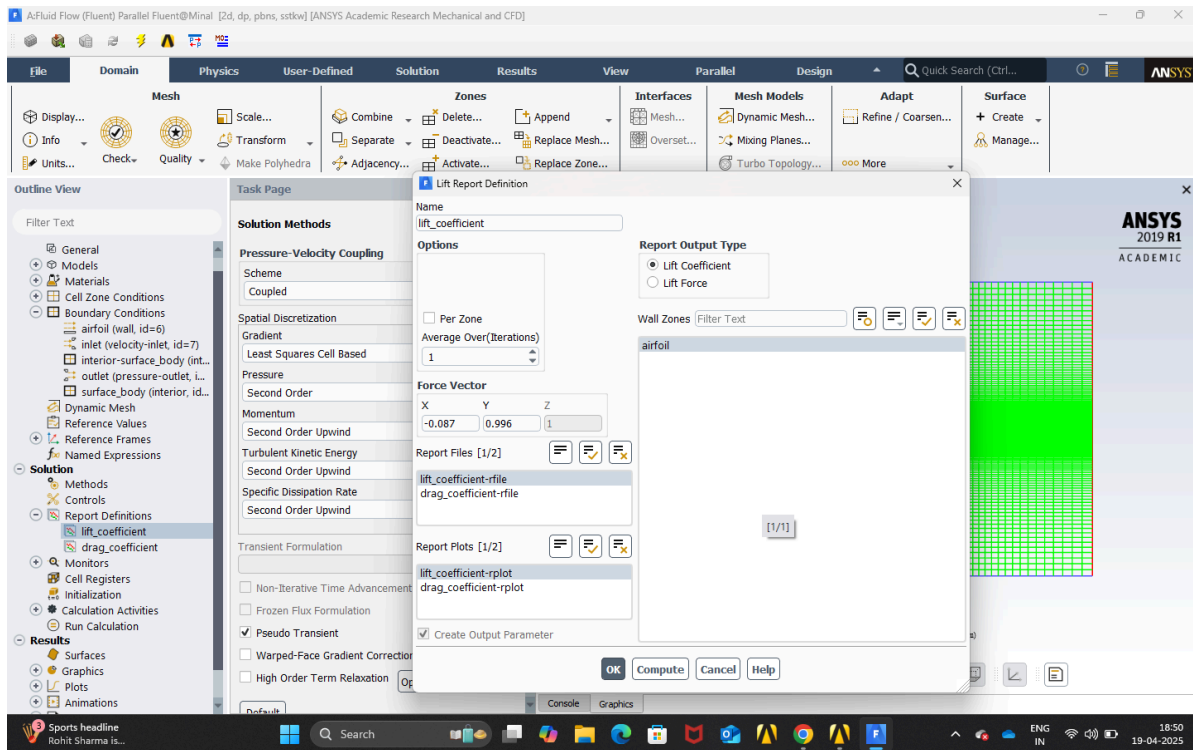


The viscosity and density values are taken at 20°C.

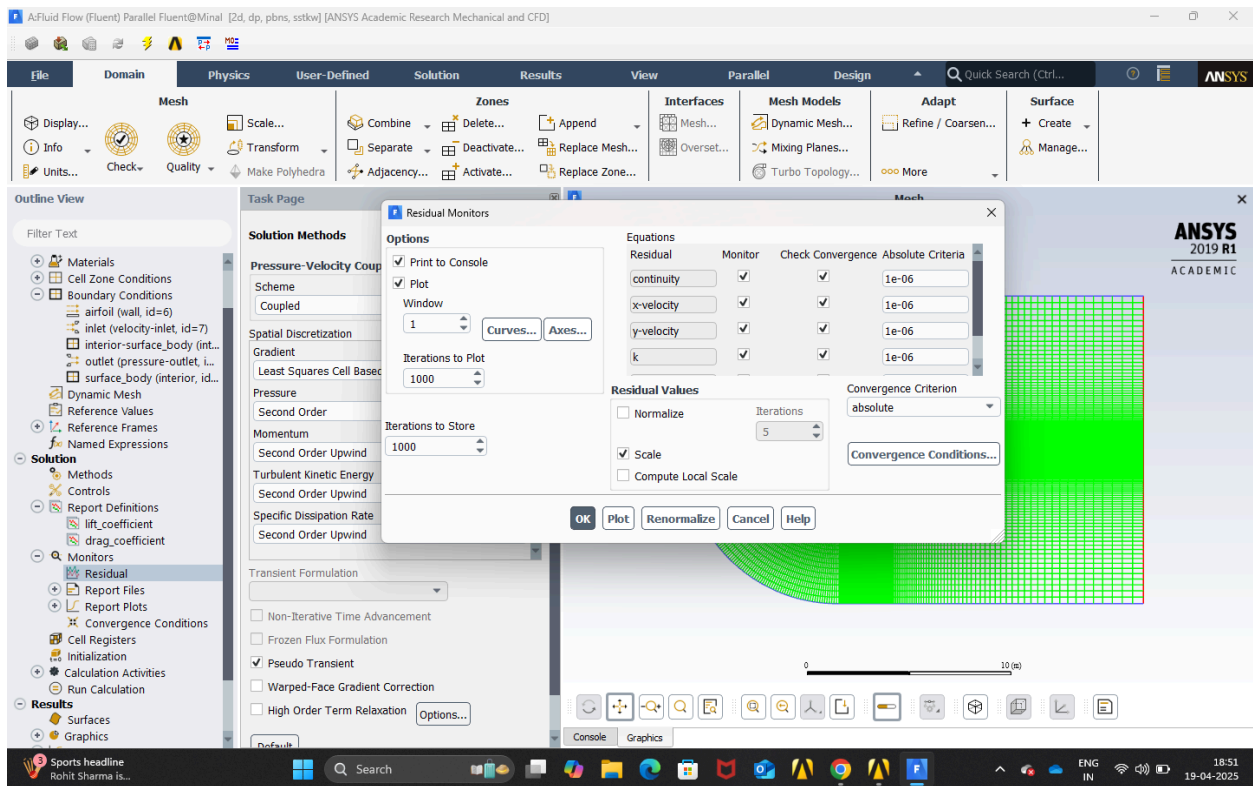
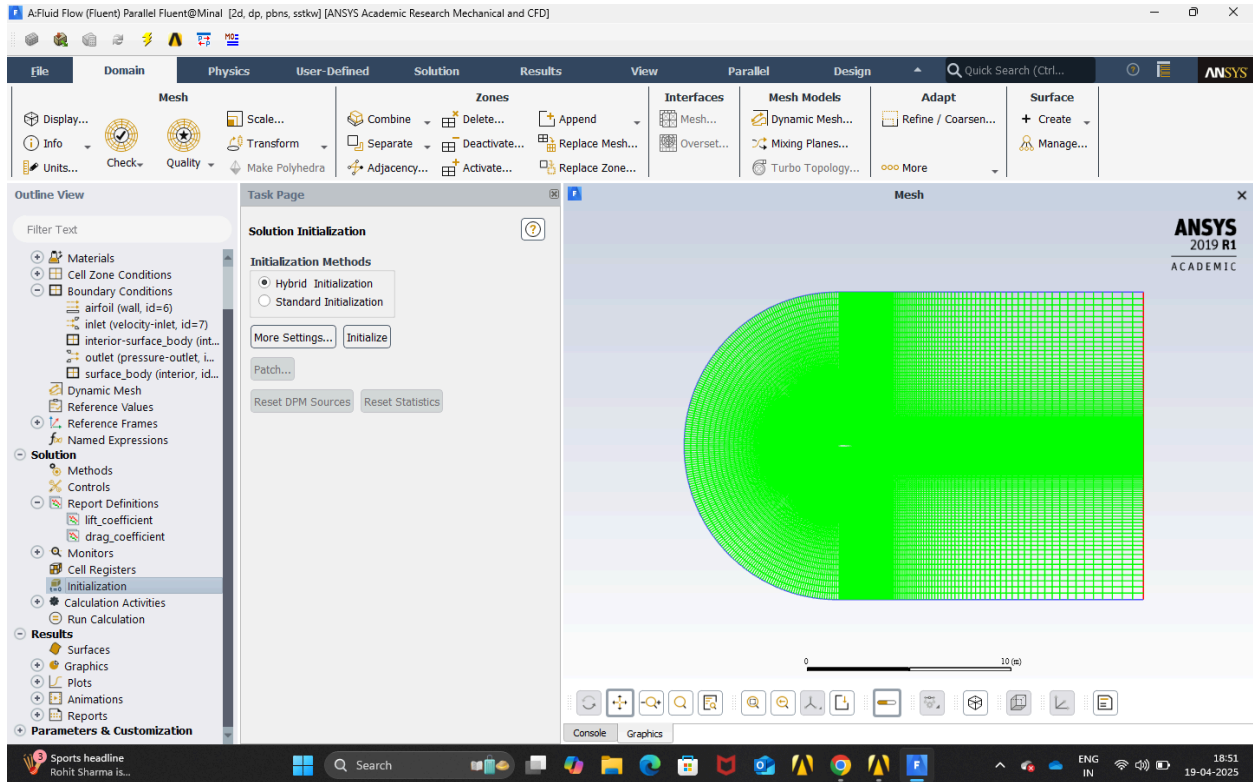
The velocities are broken into x and y components for 5° angle of attack.

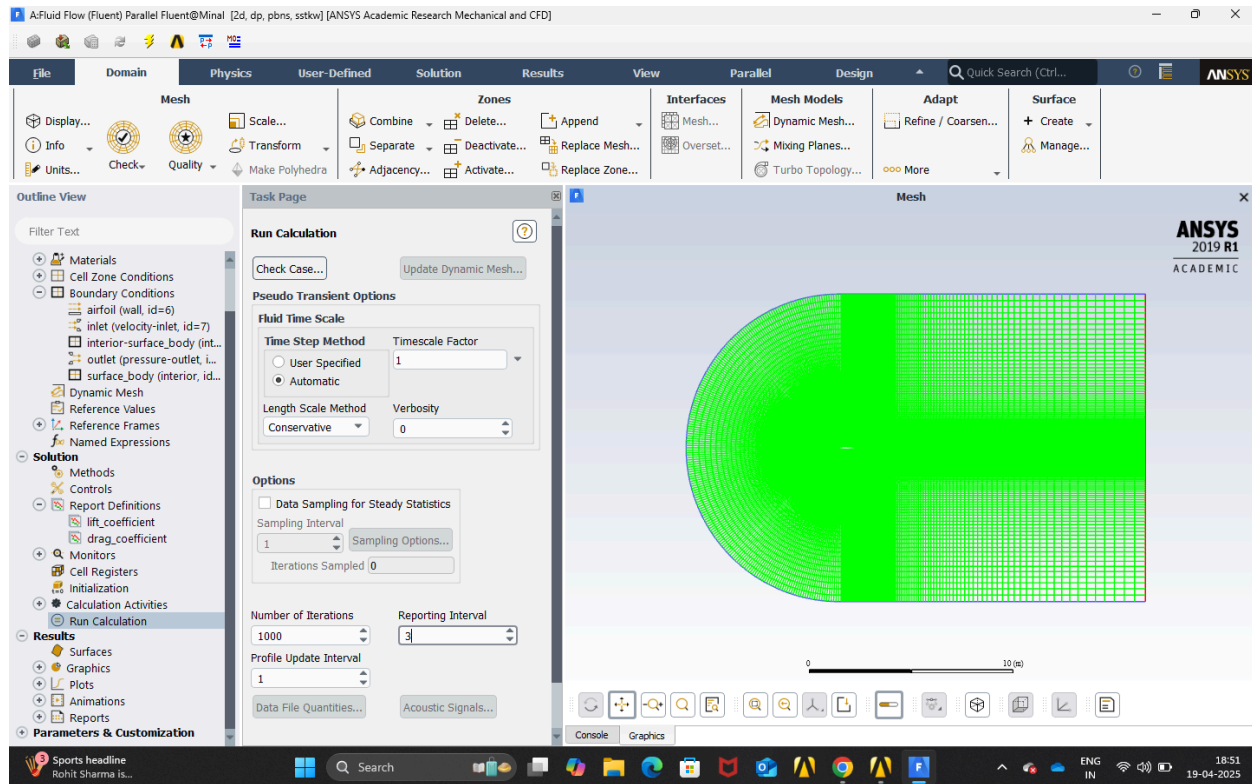






This step is done to plot the lift and drag coefficient graph





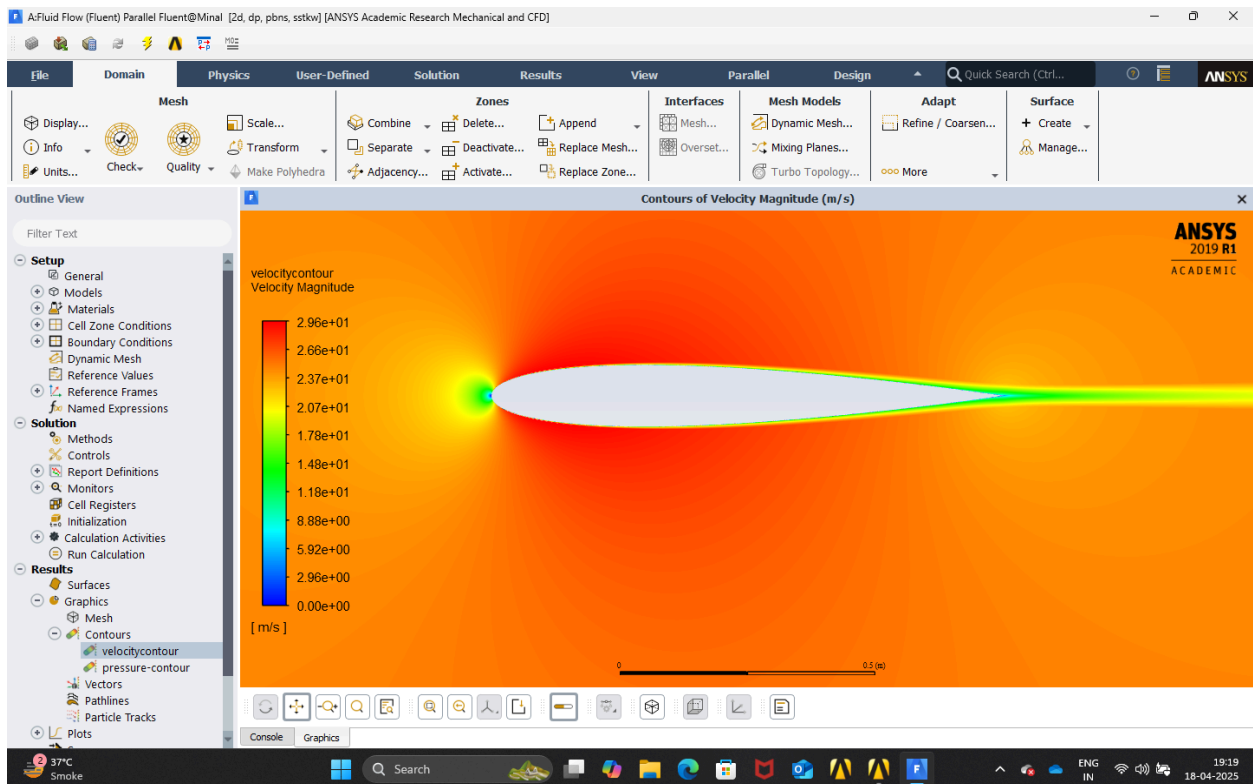
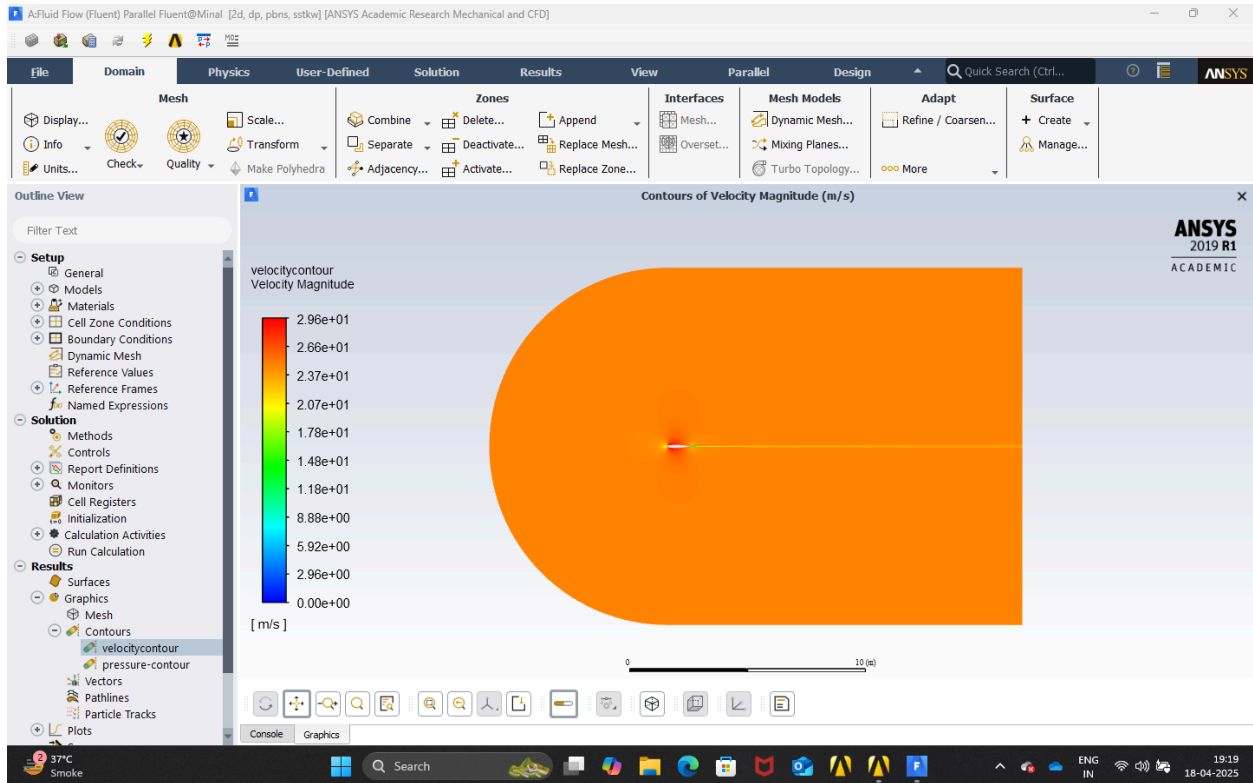
Final step is to run the calculation and the number of iterations is set 1000.

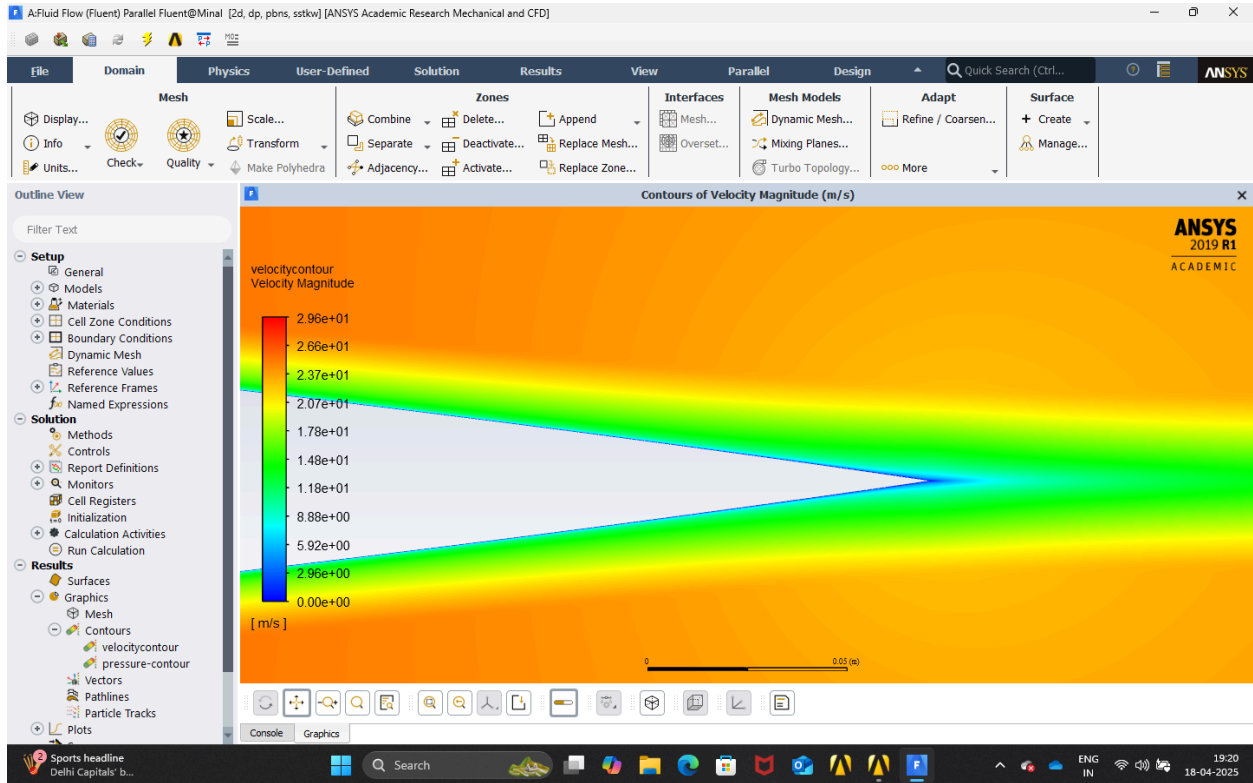
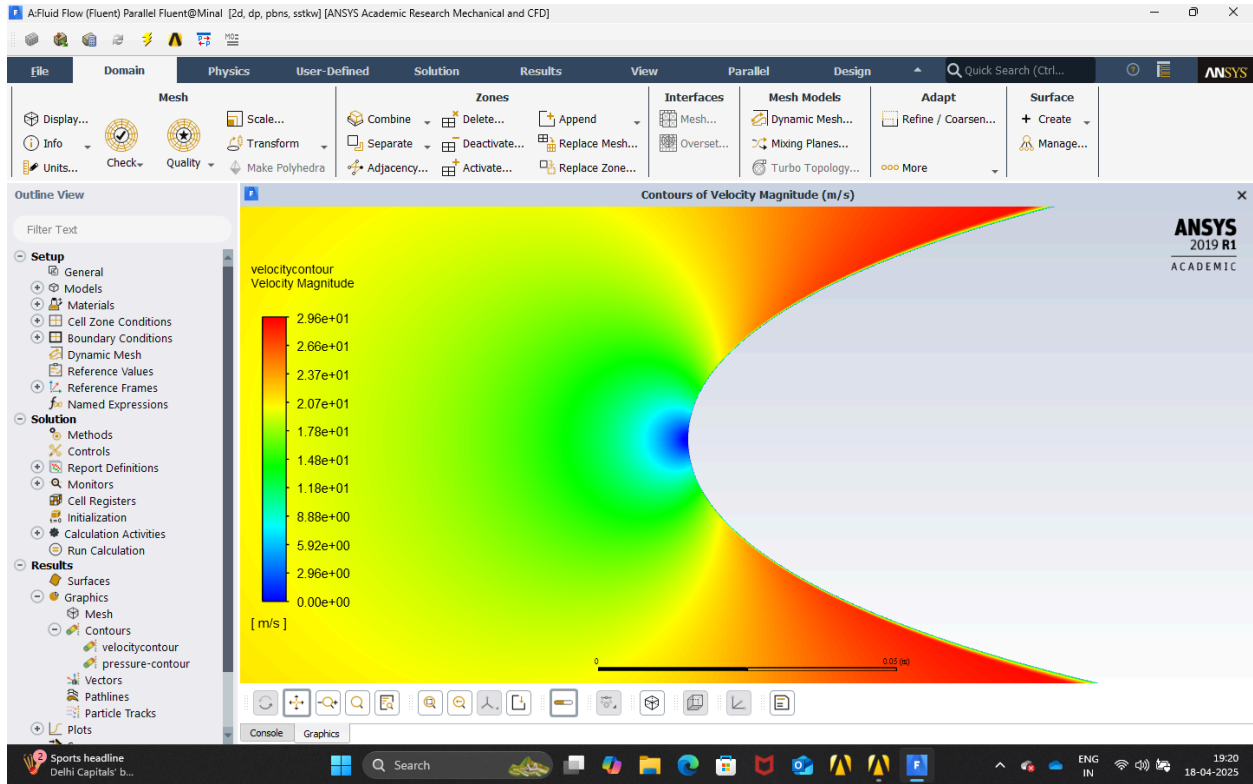
- For 0° angle of attack the only changes have to be made to the velocity components because now there will be x - component of velocity and the next change will be change to the drag and lift coefficient force vector which will now be (X - 1, Y - 0, Z - 0) for drag coefficient and (X - , Y - 1, Z - 0) for lift coefficient.
- For 5° angle of attack the change to the drag and lift coefficient force vector was made because they are measured parallel and normal to the flow directions respectively.

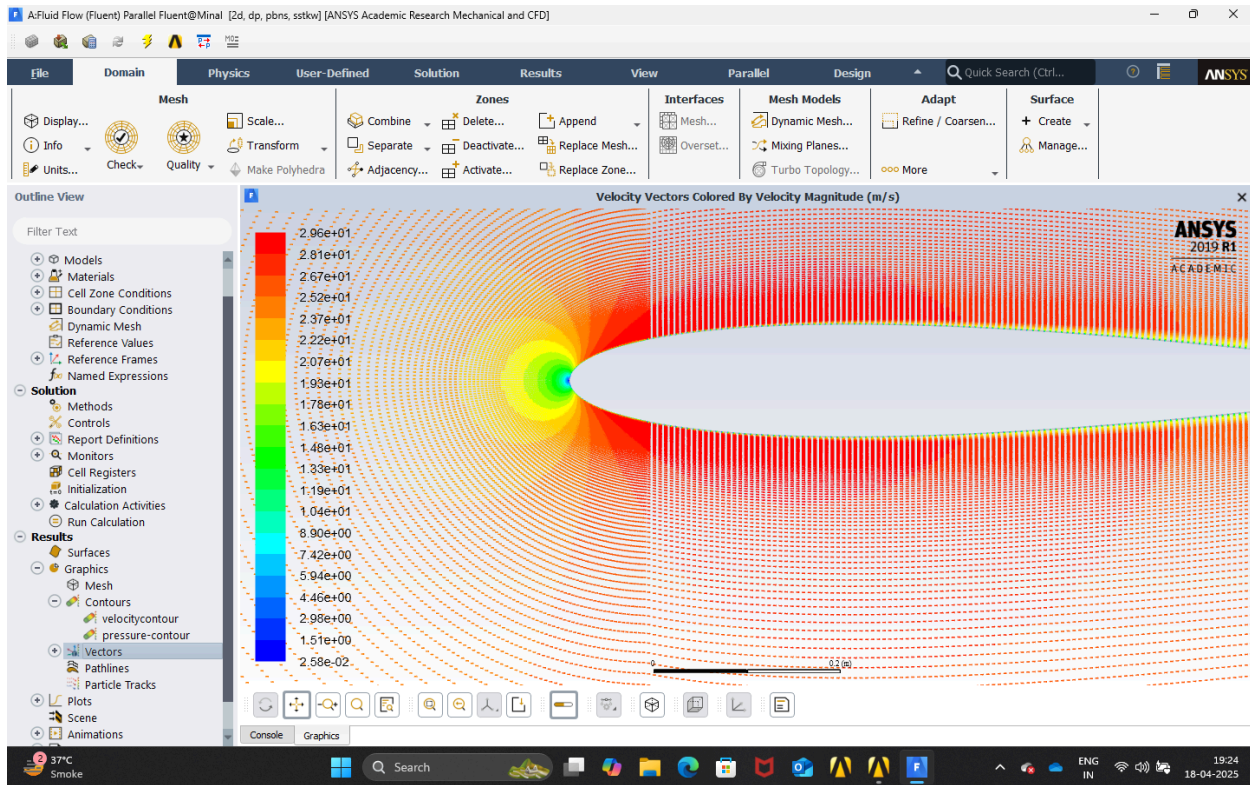
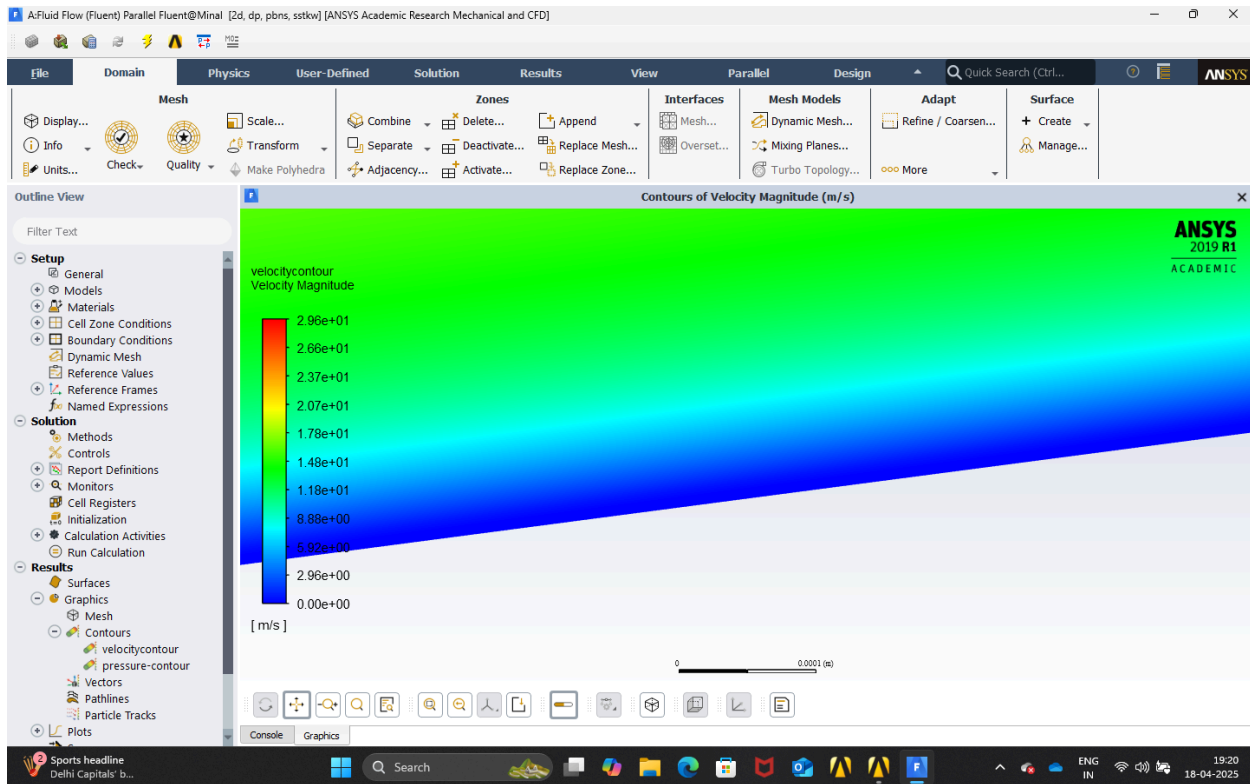
Results and Discussions

Case - 1 Angle of attack = 0 degrees

Velocity Contours



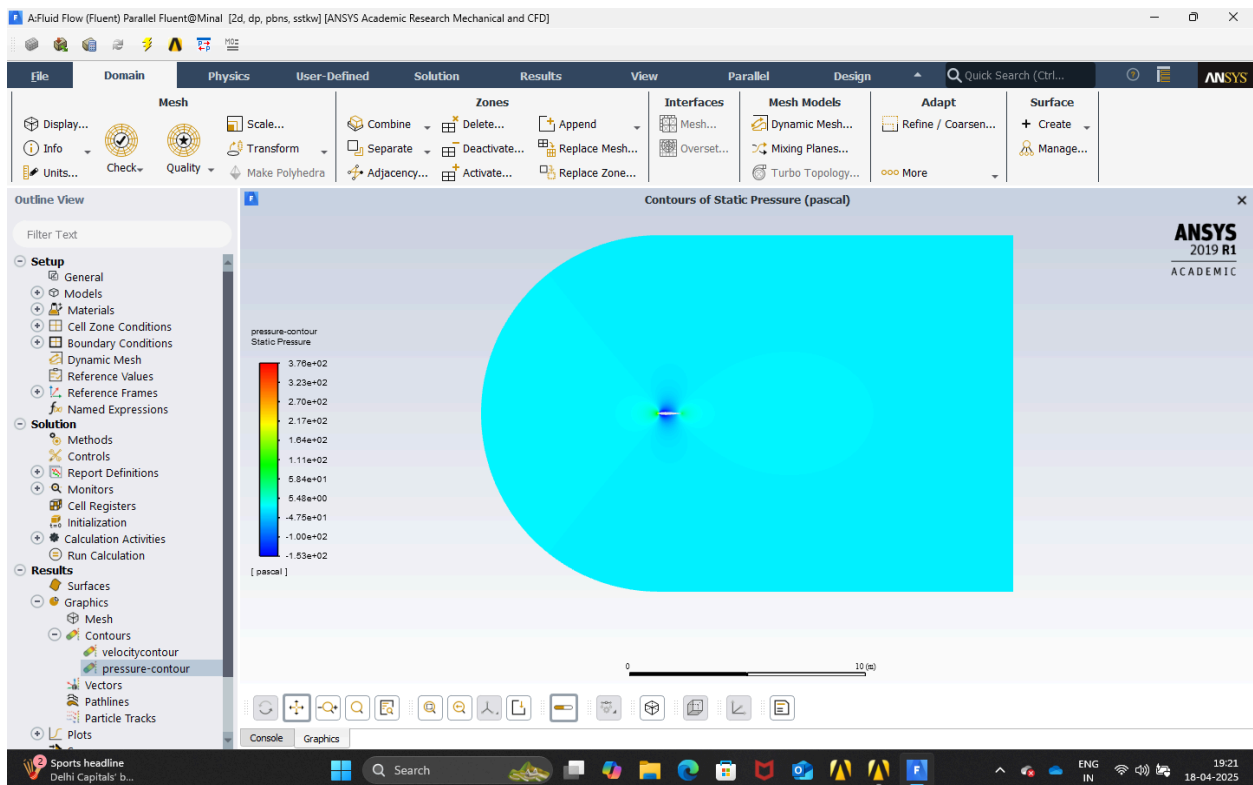


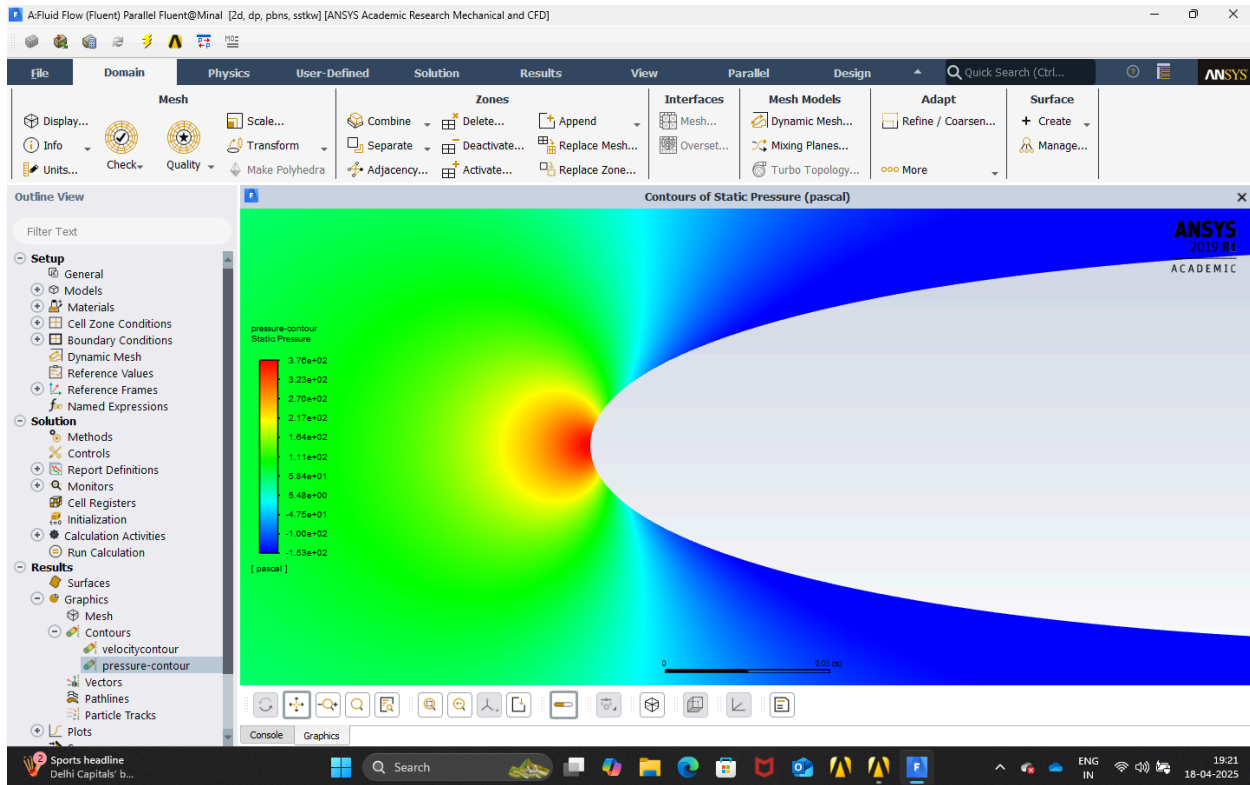
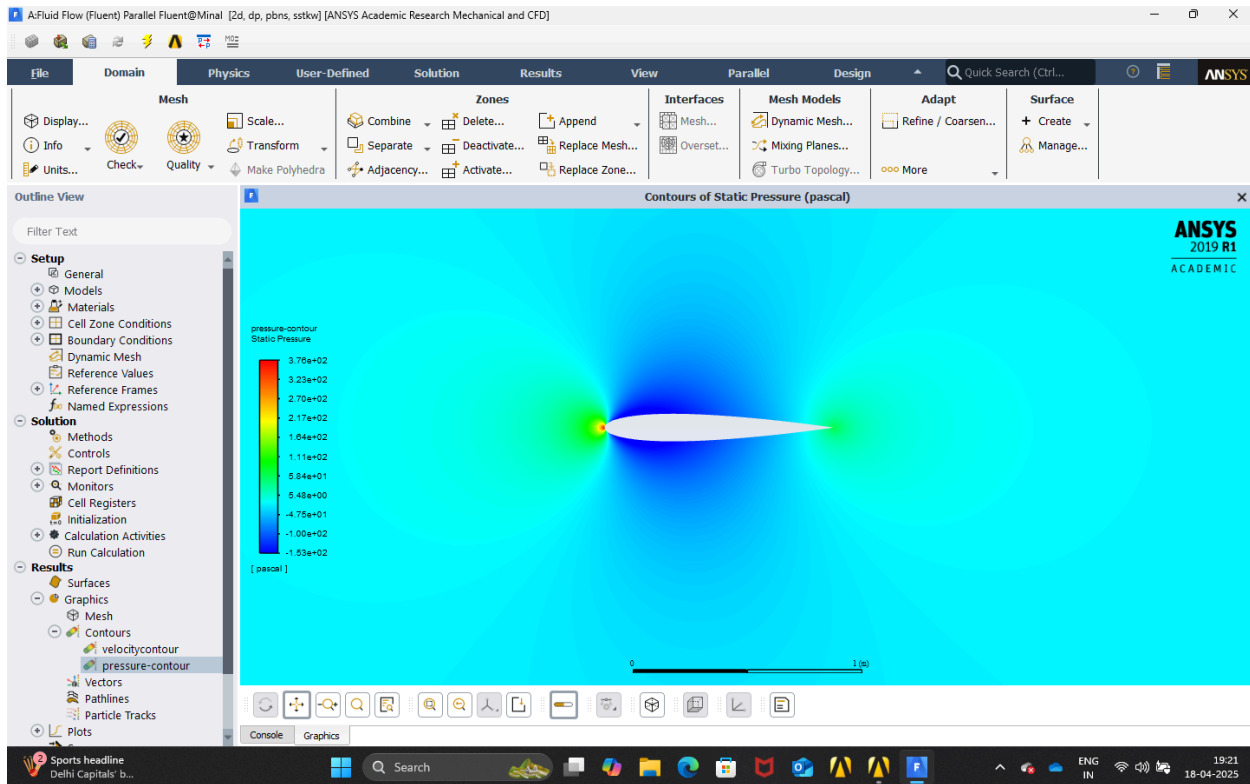


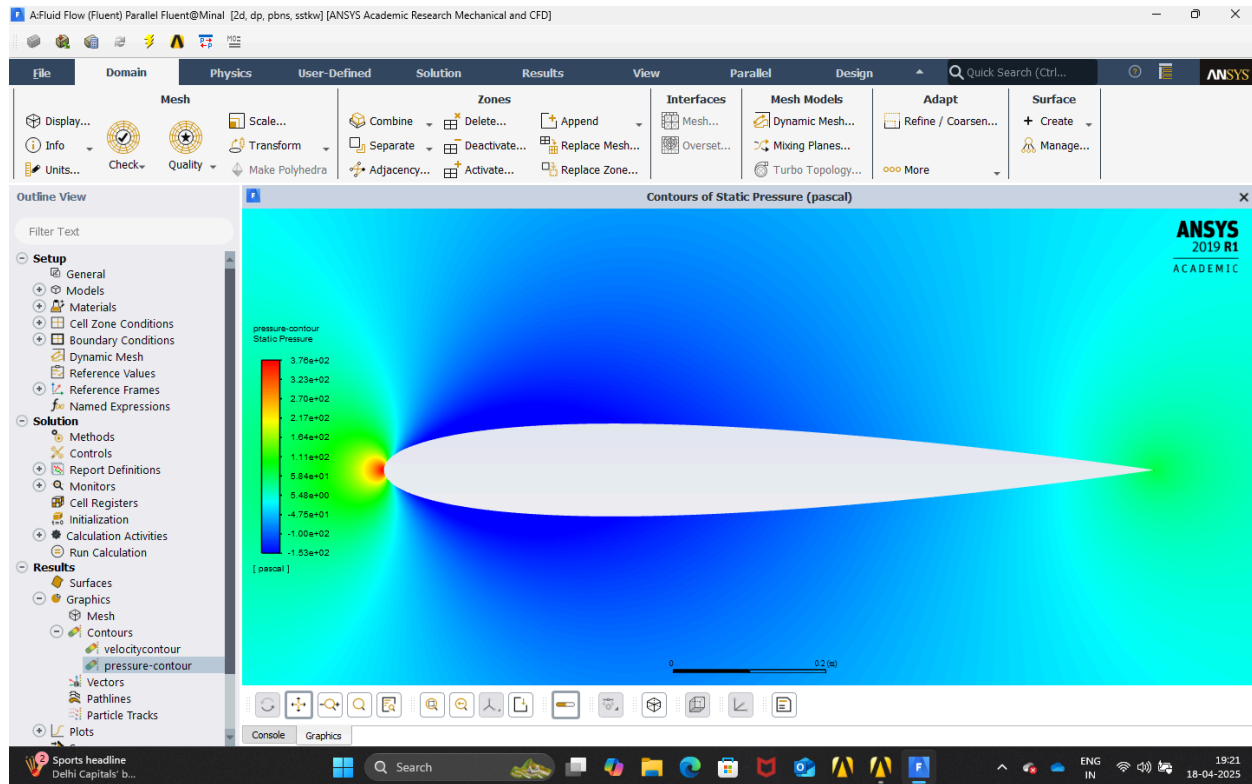
Observations

- The dark blue colour on the walls represents the no slip condition.
- The contours are symmetric as the angle of attack is zero degrees and the airfoil is symmetric.
- At the tip of the airfoil the velocity is zero indicated by dark blue showing stagnation point.
- Just above and below the walls the velocity reaches its maximum value.

Pressure Contour



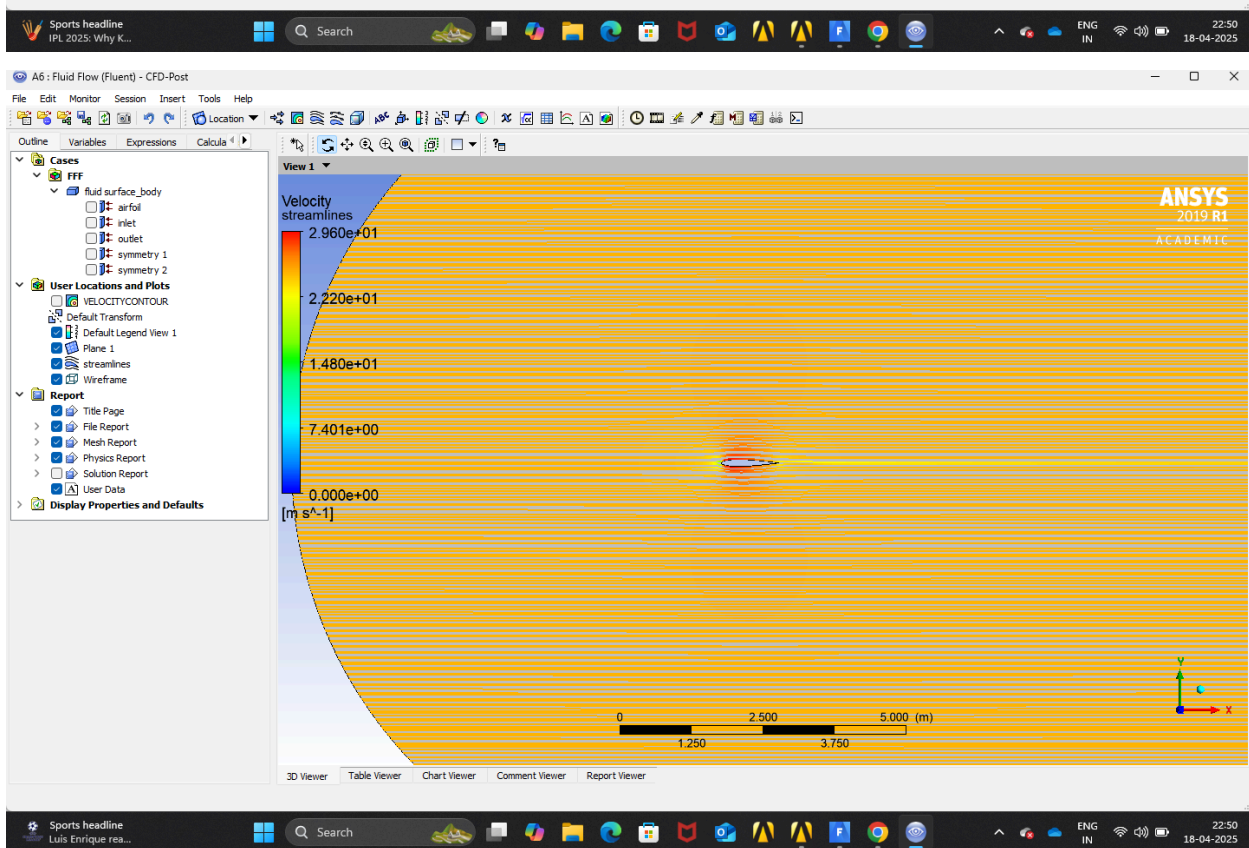
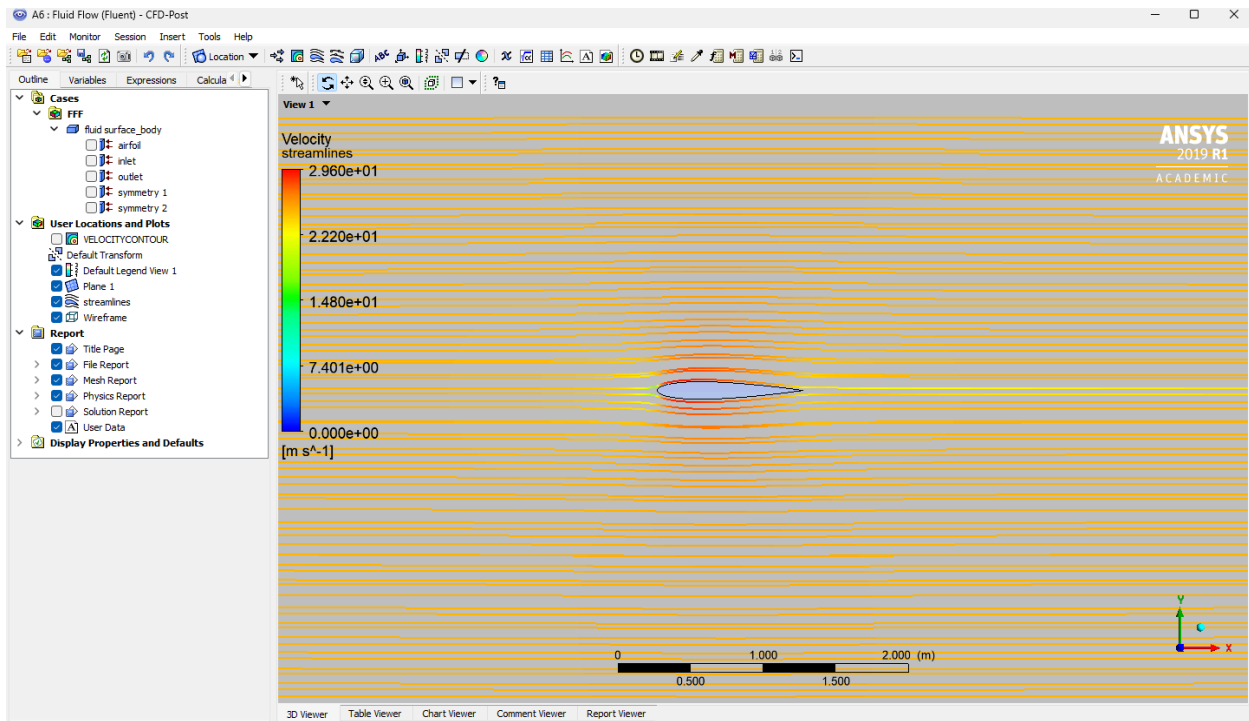




Observations

- The pressure is maximum at the wall following the no slip condition.
- The contours are symmetric as the angle of attack is zero degrees and the airfoil is symmetric.
- At the tip of the airfoil the pressure is maximum indicated by dark red showing stagnation point following Bernoulli's principle.
- Just above and below the walls the pressure reaches its minimum value also following Bernoulli's principle.

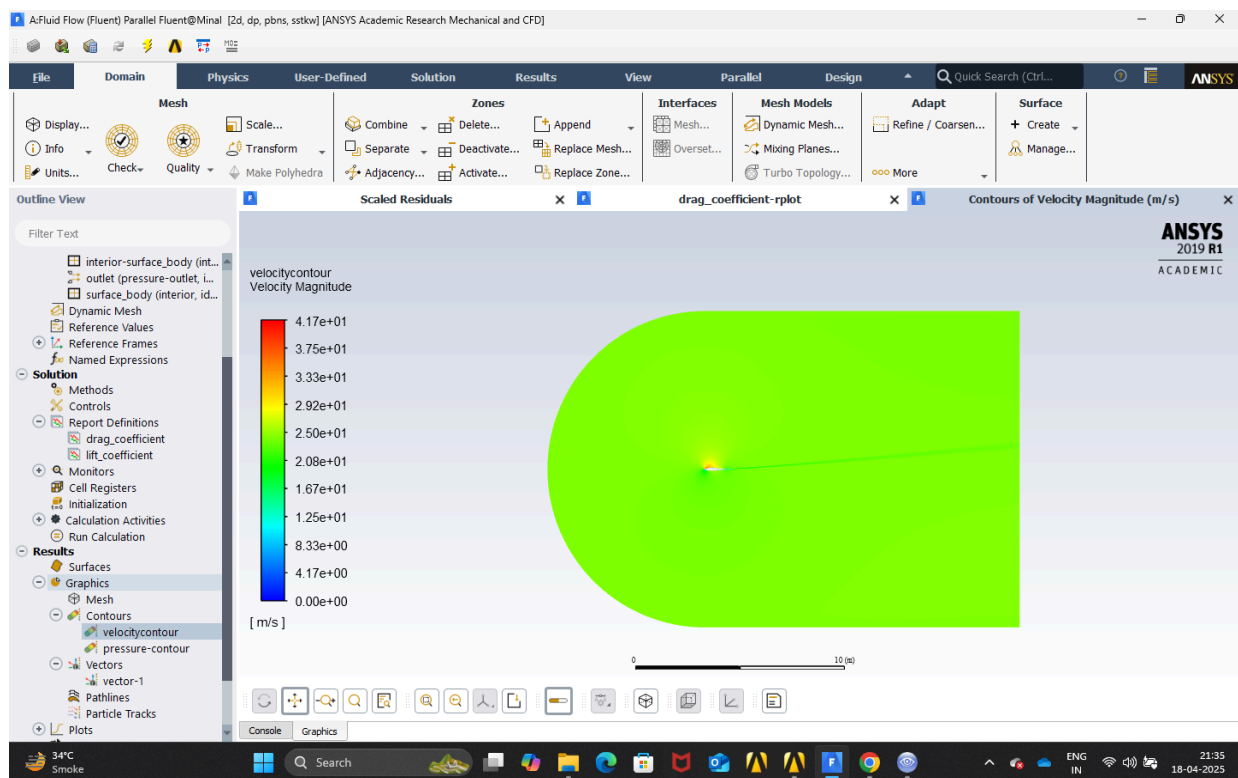
Streamlines

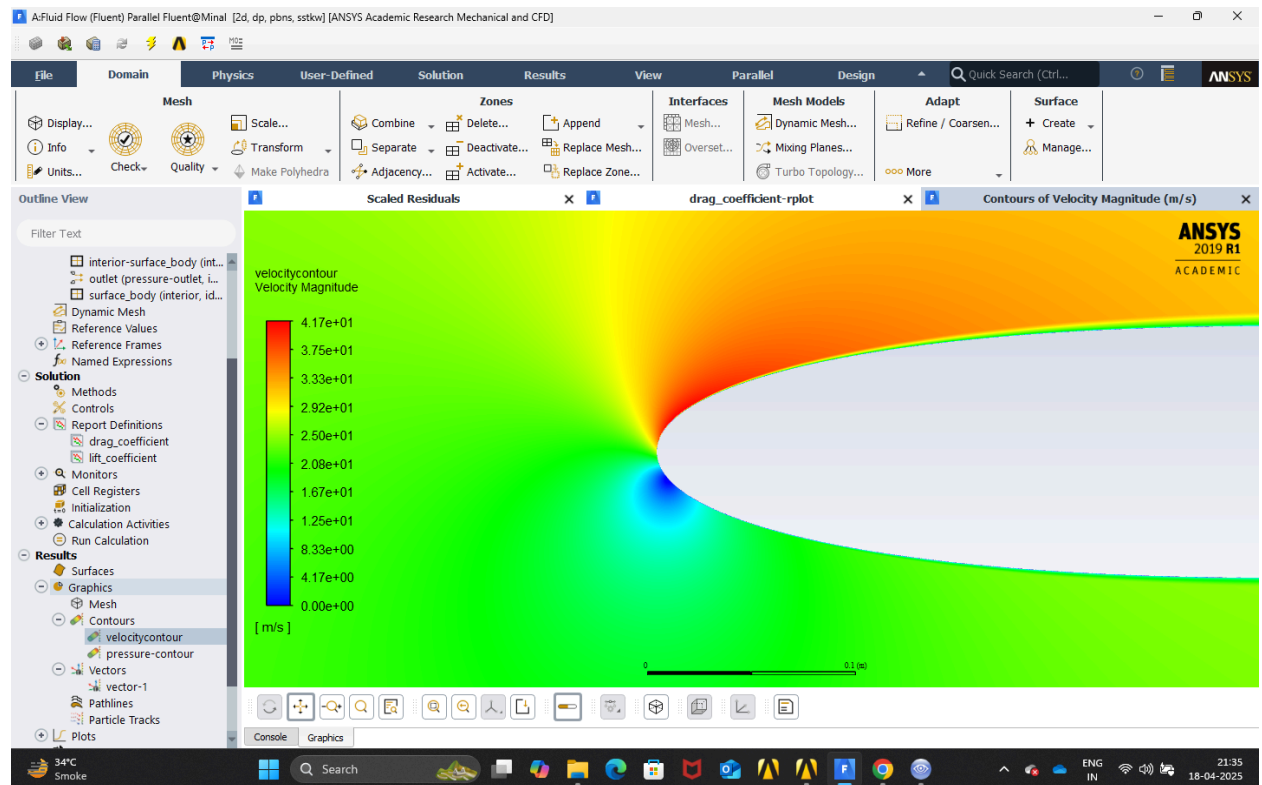
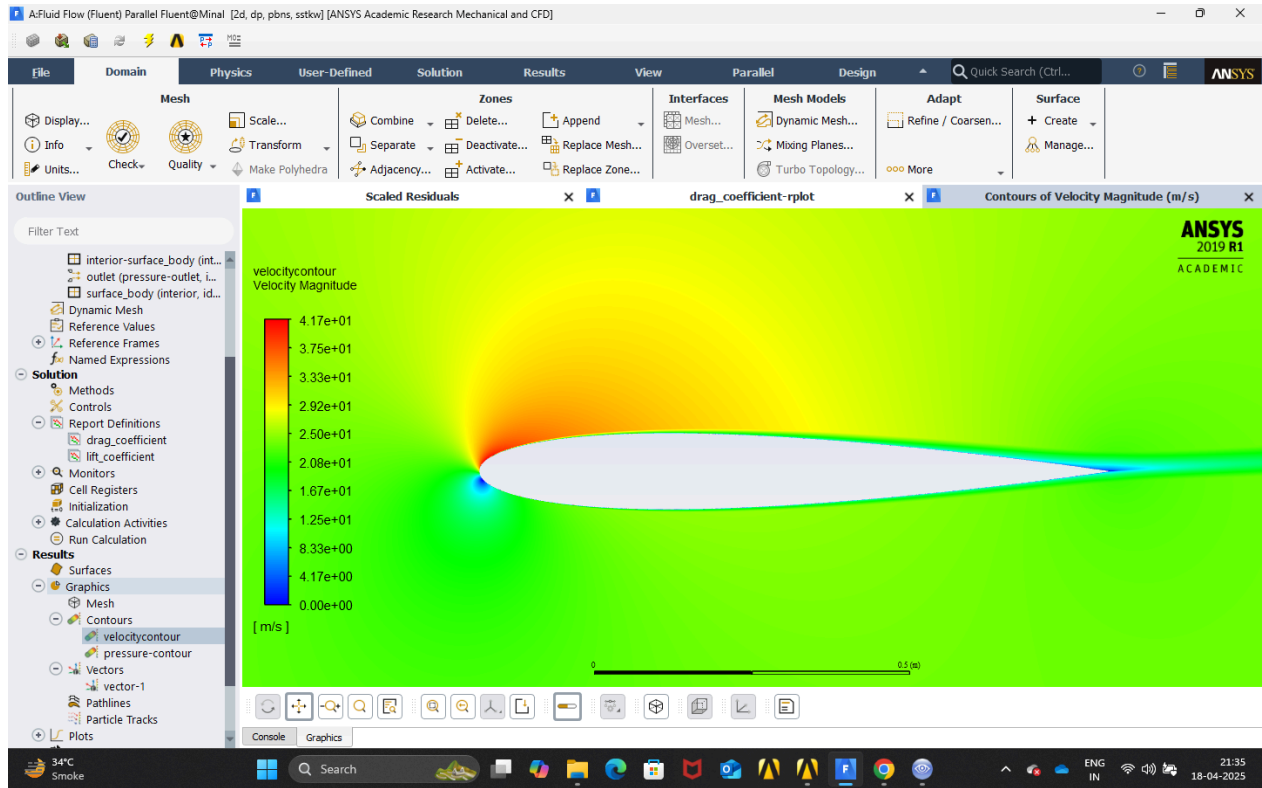


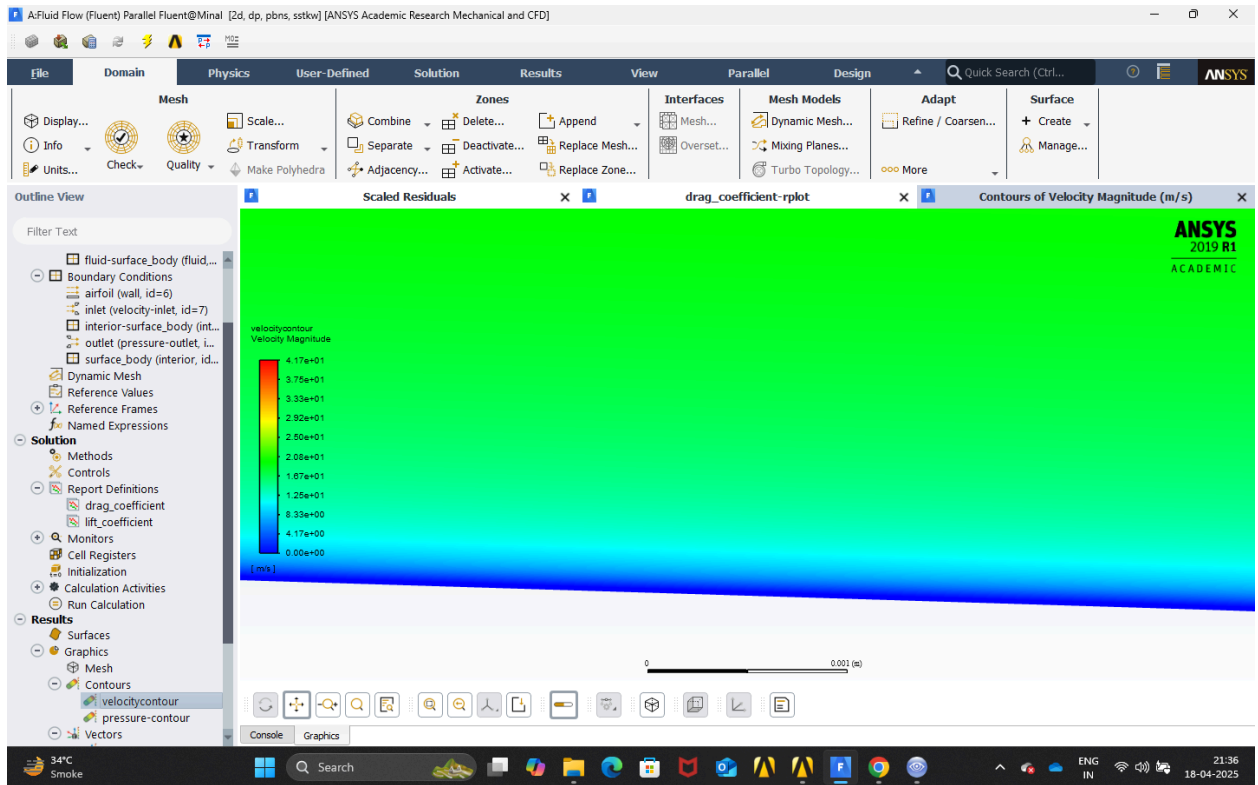
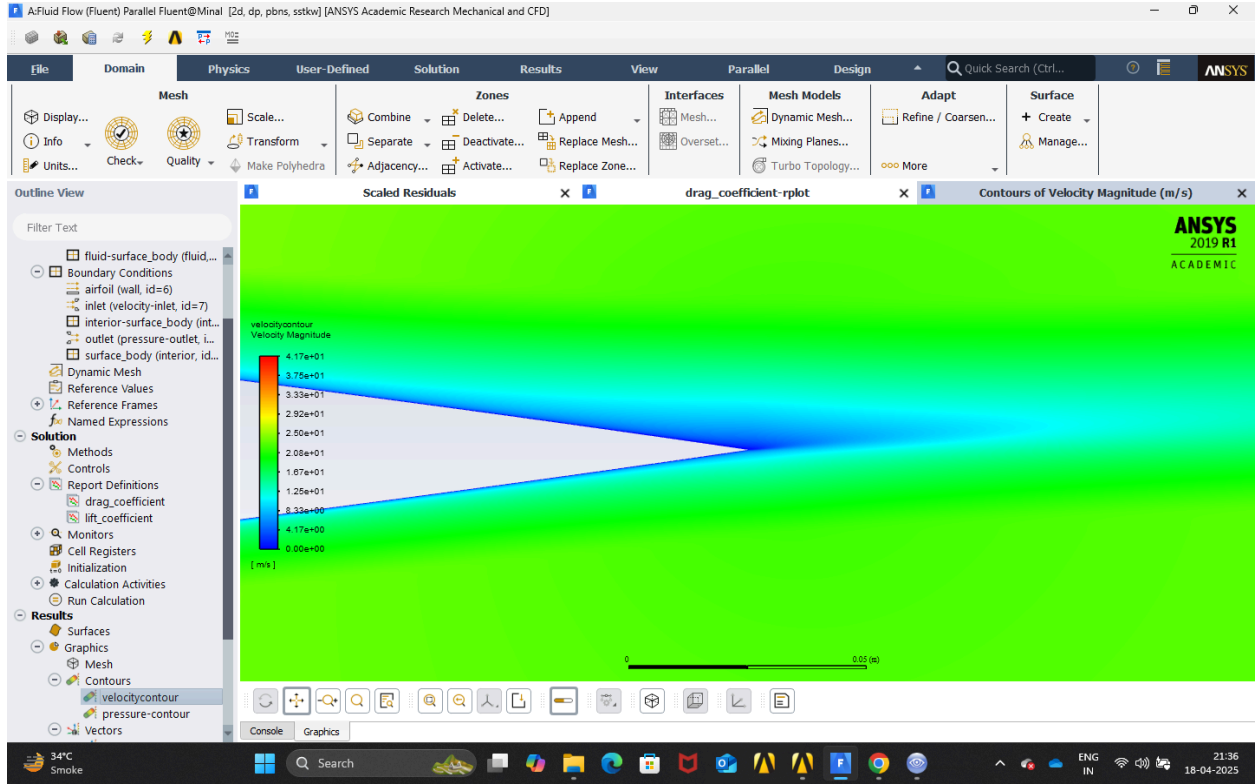
Observations

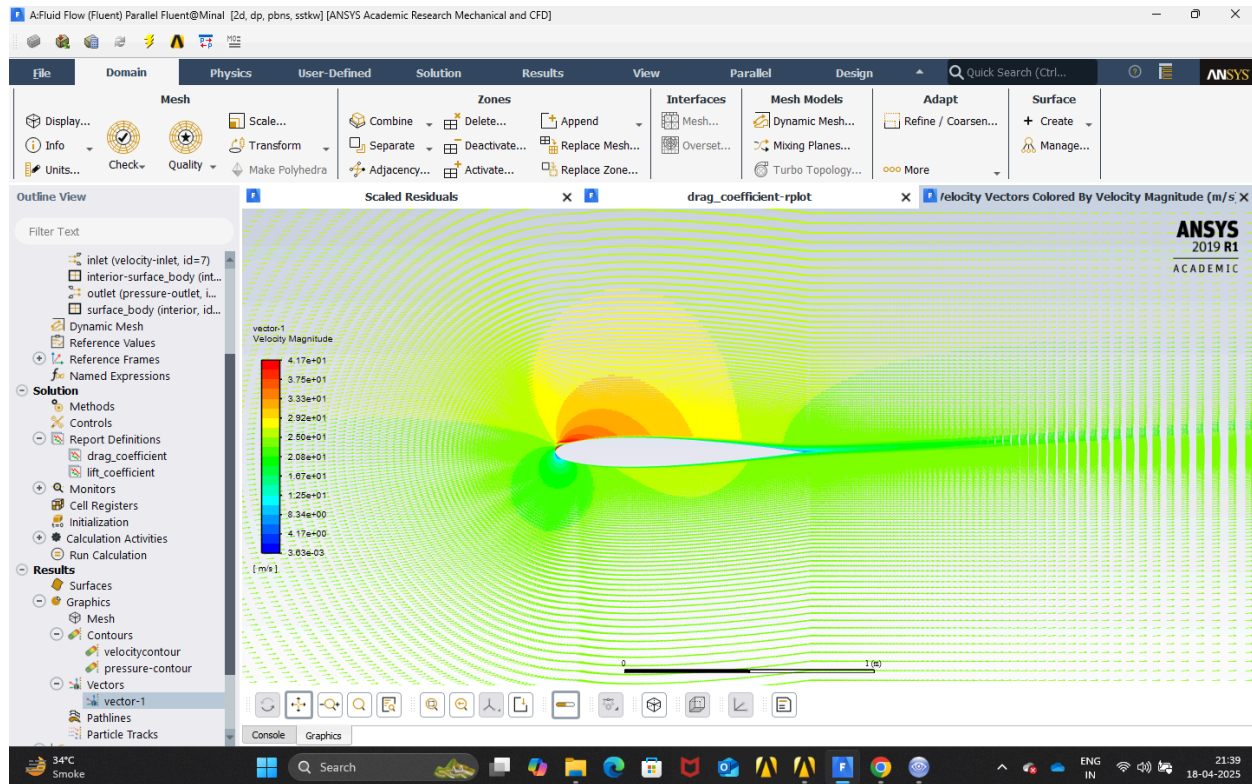
- The streamlines faraway from the airfoil are mostly a horizontal line as the flow is only in the x - direction.
- Near the airfoil we can observe the streamline flowing over it as the streamlines are curved.
- The velocity increases from the tip of the airfoil to the middle and then again decelerates and it is indicated by the streamlines getting to a color of dark red and then again going back to yellow.

Case - 2 Angle of attack = 5 degrees Velocity Contour





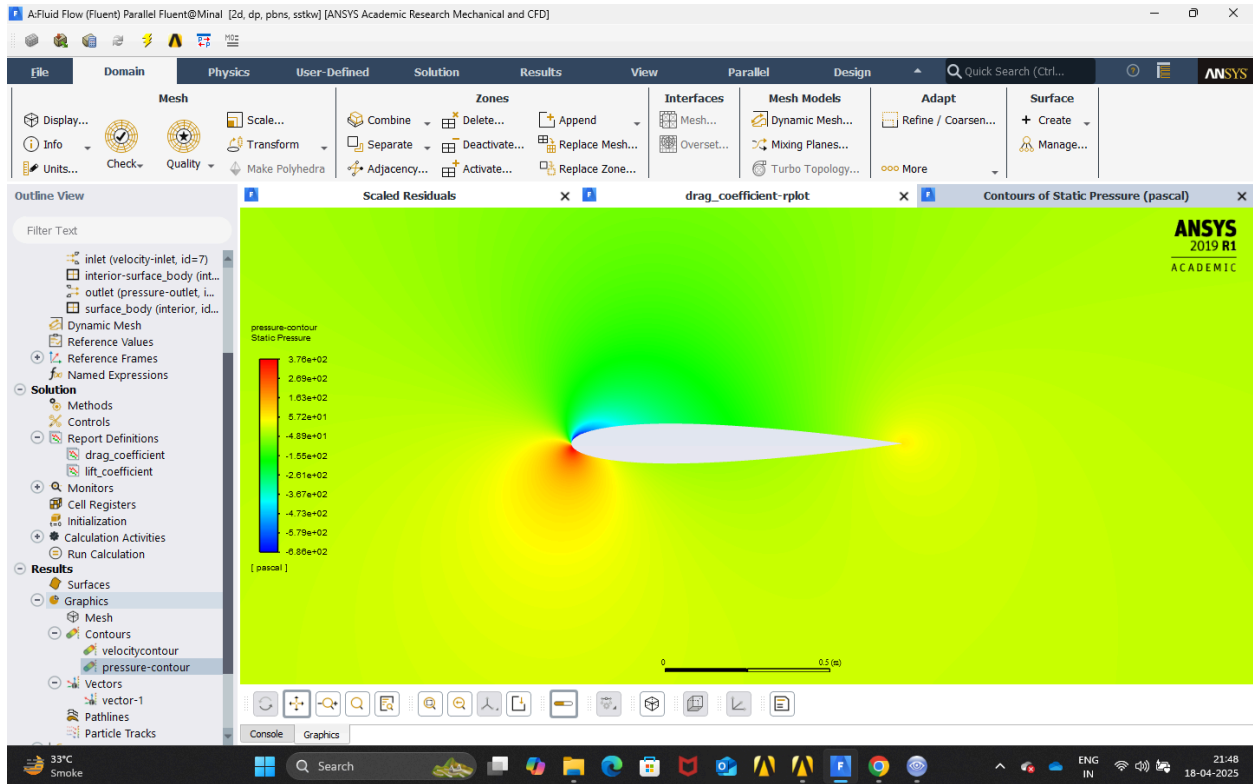
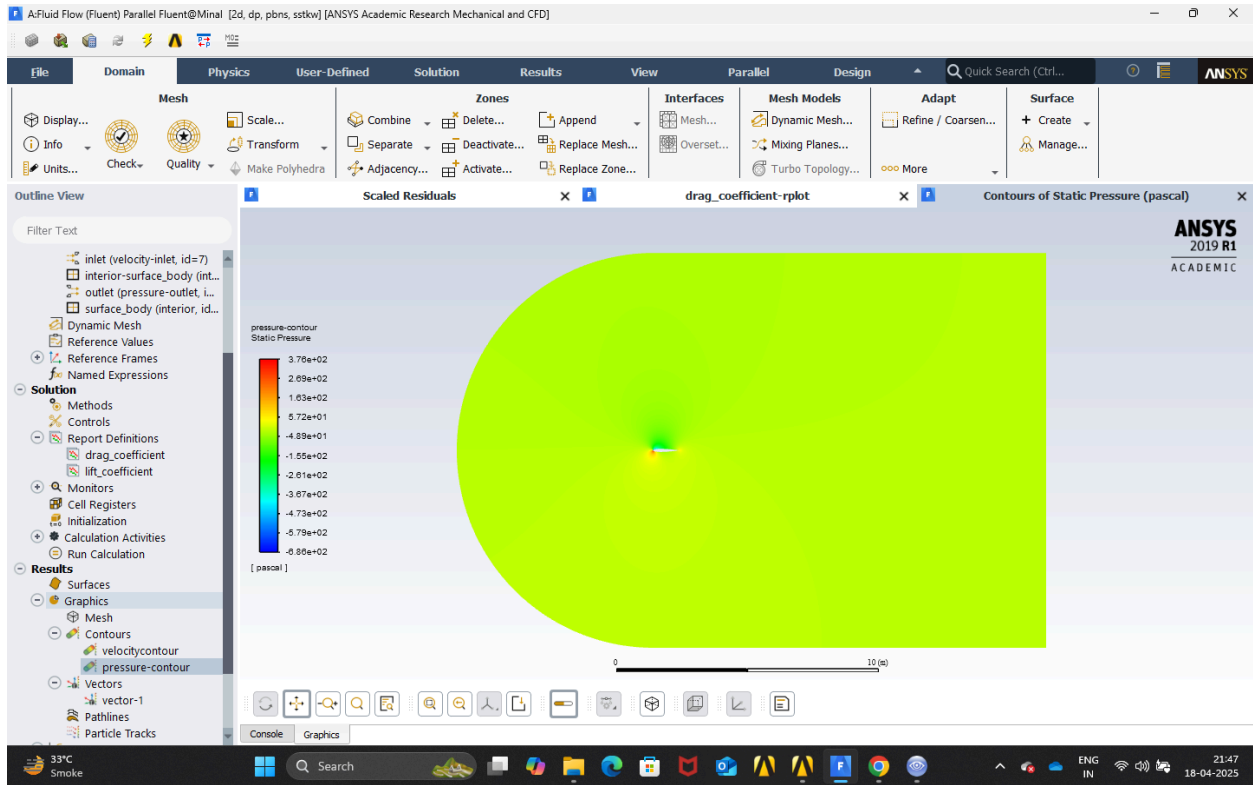


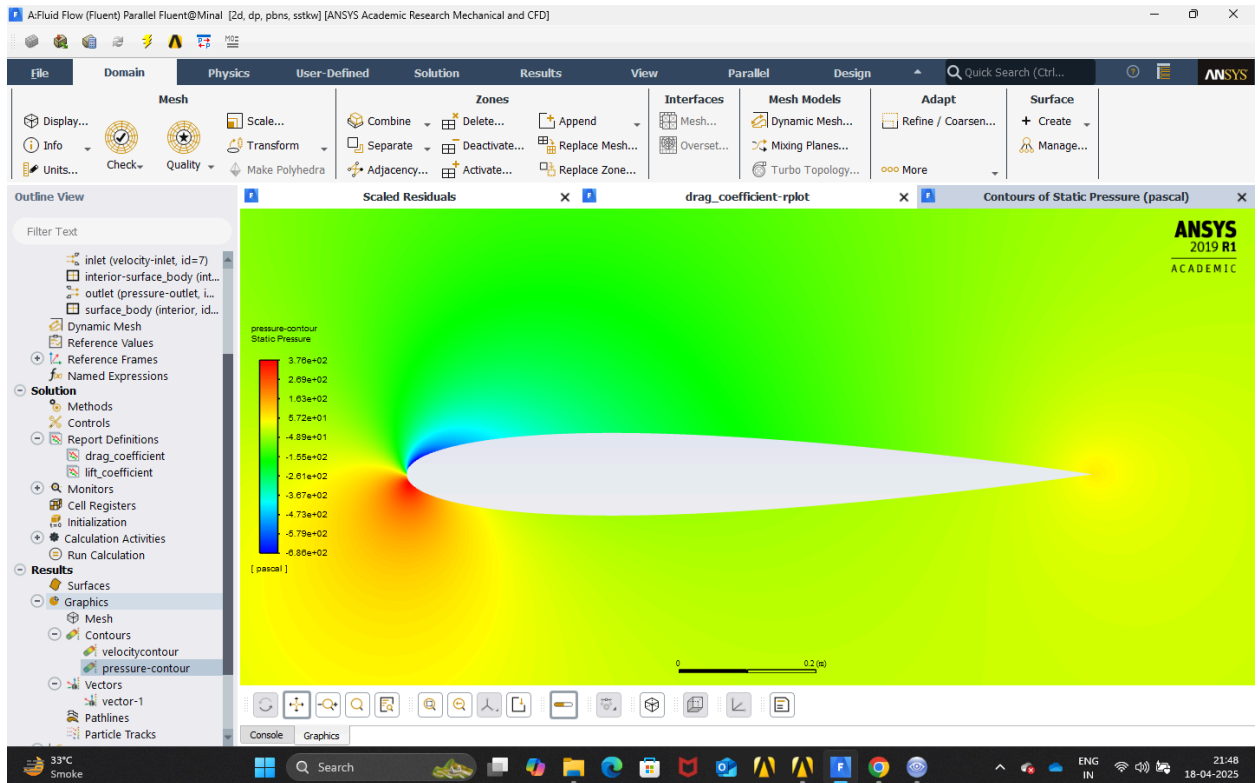
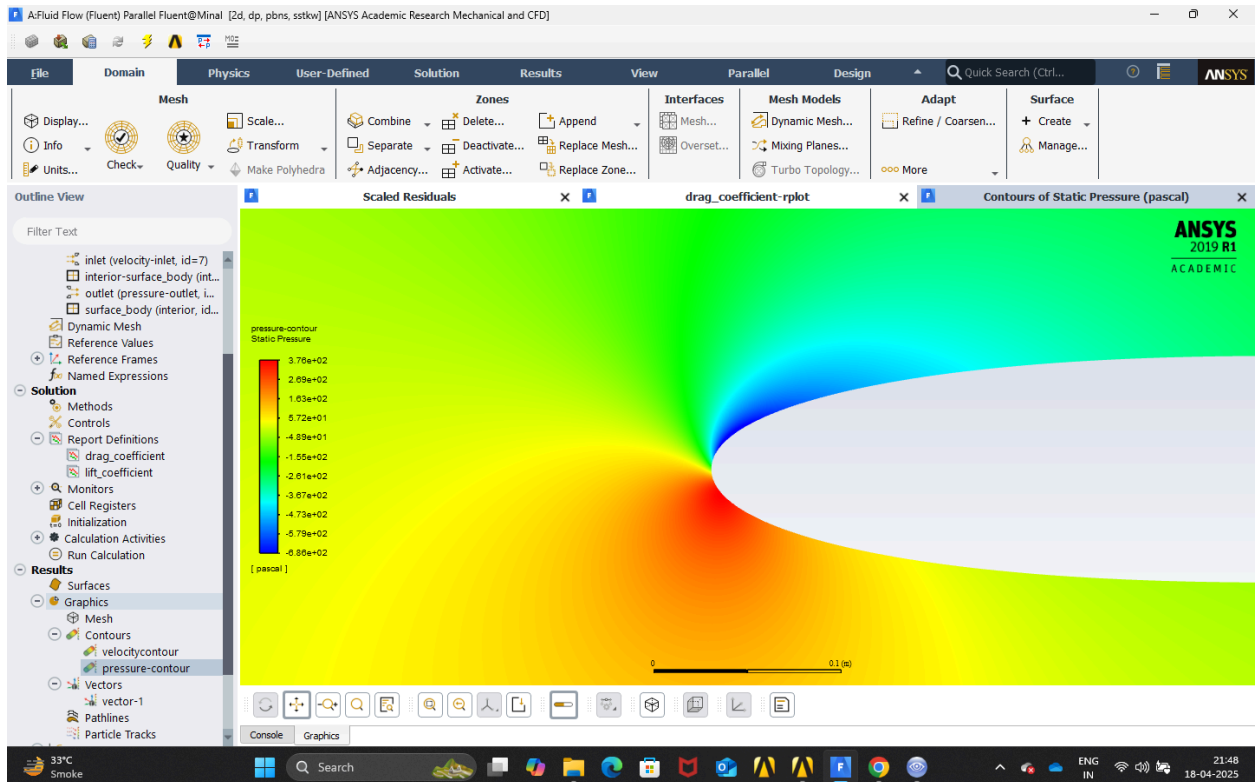


Observations

- The dark blue colour on the walls represents the no slip condition.
- The contours is not symmetric as the angle of attack is 5 degrees,
- The point at the tip of the airfoil where the velocity is zero has shifted a little down as the reason is the angle of attack is now 5 degrees so the position of stagnation point has changed.
- In general the velocity above the airfoil is more than the velocity below the airfoil which shows that the lift coefficient will be positive which indeed it is.
- The no slip condition indicated by dark blue colour is still valid

Pressure Contour

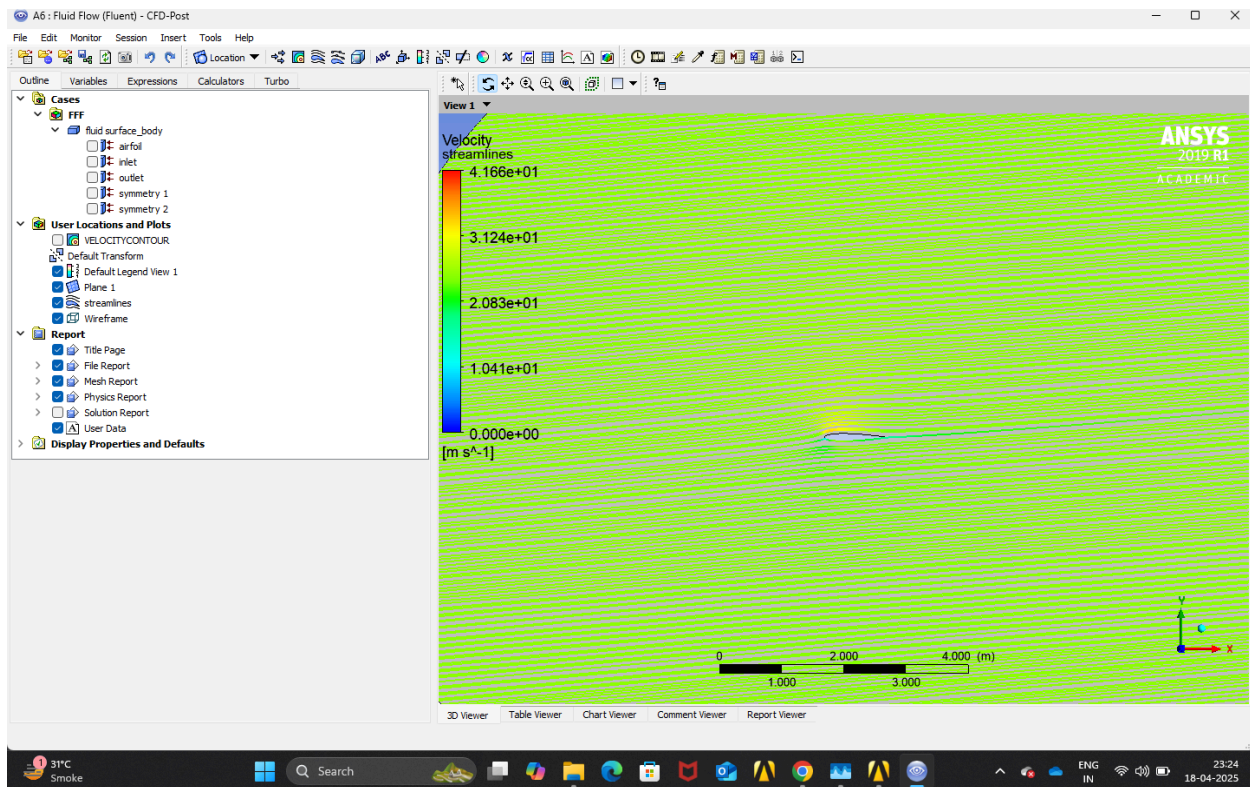


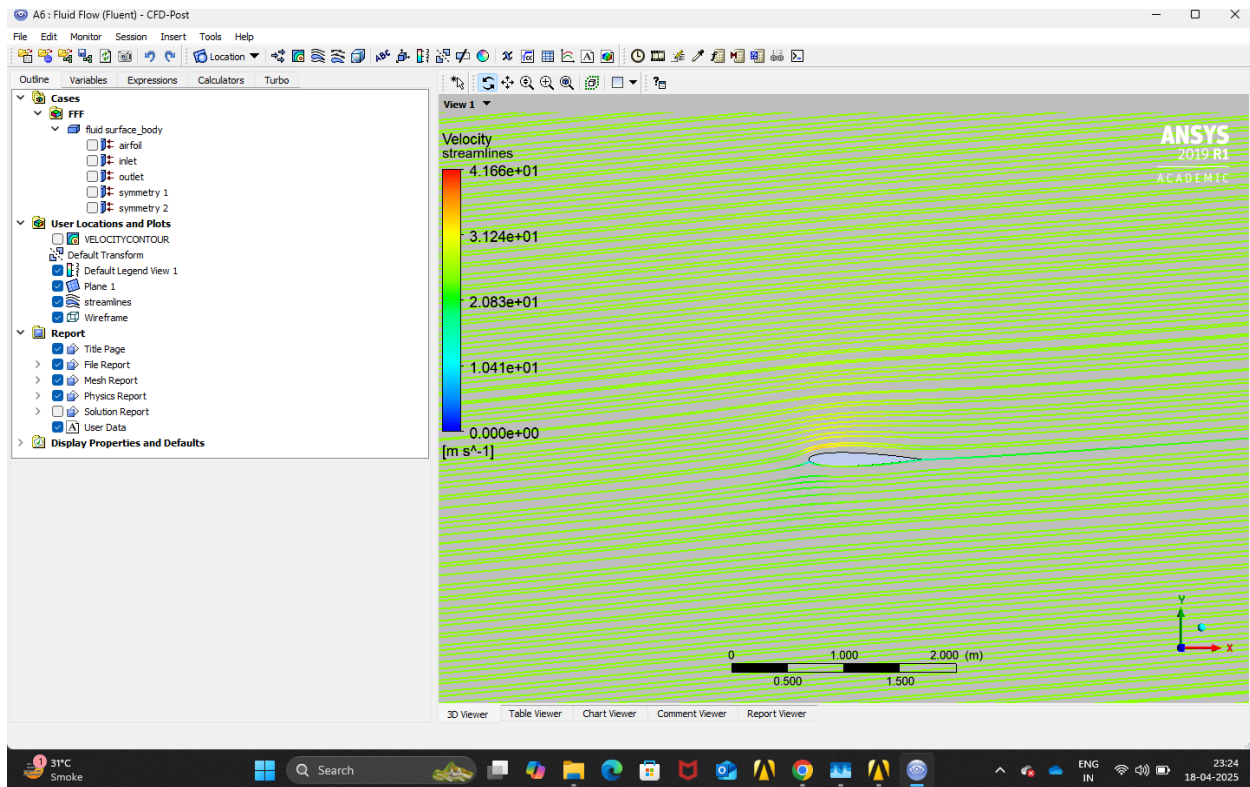


Observations

- The pressure is equal to zero at just above the tip of the airfoil.
- The pressure is maximum at just below the tip of the airfoil.
- The contours are not symmetric as the angle of attack is now 5 degrees.
- The contours are following Bernoulli's principle as in the velocity contour the point where velocity is maximum, the pressure is minimum and vice-versa.
- In general the pressure below the airfoil is greater than pressure above the airfoil signifying positive lift coefficient and lift force.

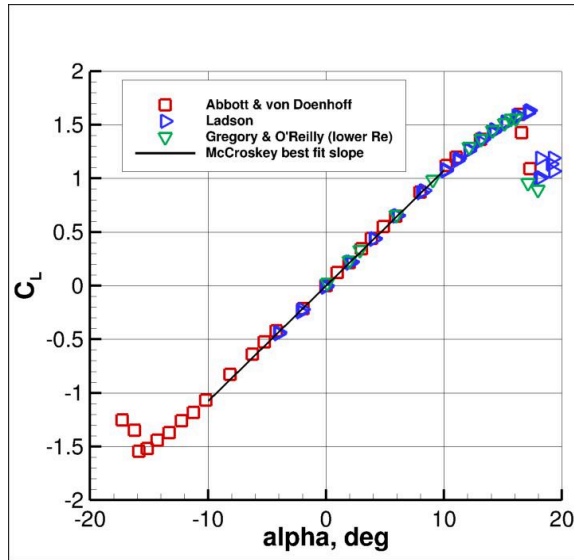
Streamlines



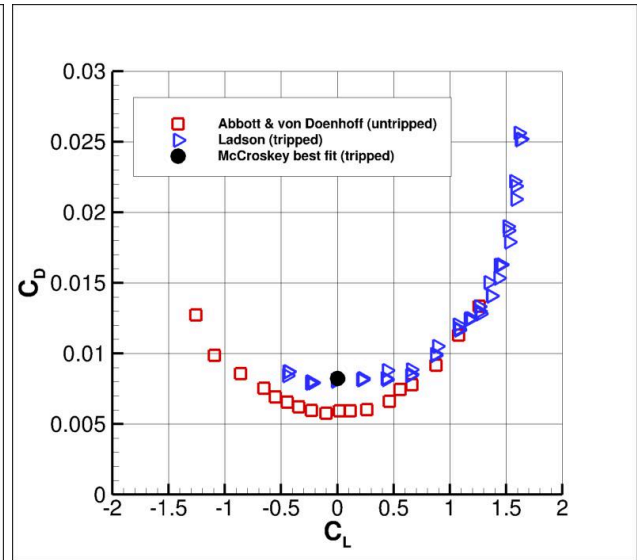


Observations

- The streamlines faraway from the airfoil are mostly parallel lines angled at 5 degrees from the horizontal because now the flow is coming at an angle of 5 degrees.
- Near the airfoil we can observe the streamline flowing over it as the streamlines are curved.
- The velocity increases from the tip of the airfoil to a point before the mid of the airfoil and then again decelerates and it is indicated by the streamlines getting to a color of yellow and then again going back to green.



Graph of AOA with Lift coefficient



Graph of Lift coefficient with Drag coefficient

Angle of Attack	Parameters	Ansys - CFD results	Literature
0°	C_D	0.010069	0.008
	C_L	6.9584 E-09	0
5°	C_D	0.011565	0.009
	C_L	0.52867	0.544

Observation of variations of lift and draft coefficients with angle of attack.

- As the angle of attack changes from 0 to 5 degrees the lift coefficient changes from a value of 6.9 E-09 to 0.52867 signifying that the lift force at zero degree will be zero and the lift force at 5 degree will not be zero which was also observed during the simulation as the lift force that I got for zero degree angle of attack was 2.6554E-6 N which is approximately equal to zero and it changed to 198.54N at an angle of attack of 5 degree, this result is also expected as the results of velocity and pressure contours explain this very well.
- As the angle of attack changes from 0 to 5 degrees the drag coefficient changes from a value of 0.010069 to 0.011565 signifying that the drag force at zero degree will be quite less and the lift force at 5 degree will be greater than 0 degrees this

result is also expected because as the angle of attack increases the area in the front of flow increases which means more flow will be restricted increasing drag force.

Conclusion

The results from the simulation match quite accurately with the literature results which suggests that the setup used is correct. The lift increased with increase in angle of attack which is also confirmed from the increased lift coefficient and the velocity and pressure contour also explains the results obtained similarly the drag coefficient also increased as the angle of attack which match the literature results it is also self explainable because the frontal area exposed to the flow increases with increase in angle of attack.

Acknowledgements

I would like to sincerely thank Mr.Ravi Teja sir for clarifying my doubts regarding Ansys Fluent and helping me understand the working CFD simulations.

References

- <https://youtu.be/3i9Ryq-m1HA?si=5BAvwQ1W2kD3JR36>
- https://www.me.psu.edu/cimbala/Learning/ANSYS/Workbench_Tutorial_Airfoil.pdf
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- https://turbmodels.larc.nasa.gov/naca0012_val.html
- <https://help.sim-flow.com/validation/naca-0012-airfoil>